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V810i S2EX Panel Handling Subsystem

ABI
Automated Board Inspection

Panel Handling Subsystem

Hardware and software that:

1. Load the panel into the system
2. Detect the panel presence inside the system
3. Unload the panel

Panel Load Sequence

Automated Board Inspection

Board Flow Direction: Left to Right

1. Close inner barrier
2. XY stage move to the loading position
3. Adjustable Rail open to the correct width (Panel Width)
4. Panel Clamp will open (Bottom clamper move down)
5. Extend PC, Board Stop Right
6. Left outer barrier will open
7. Left Panel Clear Sensor will detect the board during loading
8. Conveyor belt will turn on
9. The panel will slow down once reach the 2nd Optical PIP Sensor (Slow Sensor)
10. The panel will stop once reached the 4^{th} Mechanical PIP Sensor (Stop Sensor) and conveyor belt turn off.
11. Retract the PC, Board Stop Right
12. Panel Clamp will close (Bottom Clamper move up)
13. Left Outer barrier will close
14. The stage will move to check with the Filter Height Sensor (Applicable for High Magnification)
15. Open inner barrier

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Panel Load Sequence (with SMEMA)

Board Flow Direction: Left to Right

1. Close inner barrier
2. XY stage move to the loading position
3. Adjustable Rail open to the correct width (Panel Width)
4. Panel Clamp will open (Bottom clamper move down)
5. Extend PC, Board Stop Right
6. Waiting for the signal from the SMEMA
7. Left Outer barrier will open
8. Left Panel Clear Sensor will detect the board during loading
9. Conveyor belt will turn on
10. The panel will slow down once reach the 2nd Optical PIP Sensor (Slow Sensor)
11. The panel will stop once reached the 4^{th} Mechanical PIP (Stop Sensor) and conveyor belt turn off.
12. Retract the PC, Board Stop Right
13. Panel Clamp will close (Bottom Clamper move up)
14. Outer barrier will close
15. The stage will move to check with the Filter Height Sensor (Applicable for High Magnification)
16. Open inner barrier

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Panel Inspection Sequence (>16.4 inch)

Board Flow Direction: Left to Right

After finish inspecting 1st region of the board, reposition for the 2nd region inspection

1. Move the stage to the right outer barrier
2. Move down the bottom clamper to un-clamp the panel
3. Turn on the conveyor belts (Left to Right)
4. Detect the panel at the right panel clear sensor
5. Turn off the conveyor belts
6. Extend the PC, Board Stop Left
7. Reverse the conveyor belts direction (Right to Left)
8. The panel will slow down once reached the 3rd Optical PIP Sensor (Slow Sensor)
9. The panel will stop once reach the 1st Mechanical PIP Sensor (Stop Sensor) and conveyor belt turn off
10. Retract the PC, Board Stop Left.
11. Move up the bottom clamper to clamp the panel.
12. Continue 2nd region inspection

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Panel Unload Sequence

1. Close the inner barrier
2. XY Stage move to the unloading position
3. Move the clamper down to un-clamp the panel
4. Turn on the conveyor belt
5. Move the panel to the unloading side Panel Clear Sensor
6. Panel Clear Sensor detect the Panel
7. Turn off the conveyor belt
8. Outer barrier opens
9. Waiting for the panel to be removed from the system
10. Panel Clear Sensor become clear
11. Outer barrier close

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Panel Unload Sequence (with SMEMA)

1. Close the inner barrier
2. XY Stage move to the unloading position
3. Move the clamper down to un-clamp the panel
4. Turn on the conveyor belt
5. Move the panel to the unloading side Panel Clear Sensor
6. Panel Clear Sensor detect the Panel
7. Turn off the conveyor belt
8. Outer barrier opens
9. SMEMA will detect the board
10. Panel Clear Sensor status become clear after unload the board.
11. Outer Barrier Close.

GUI Interface - Digital I/O tab

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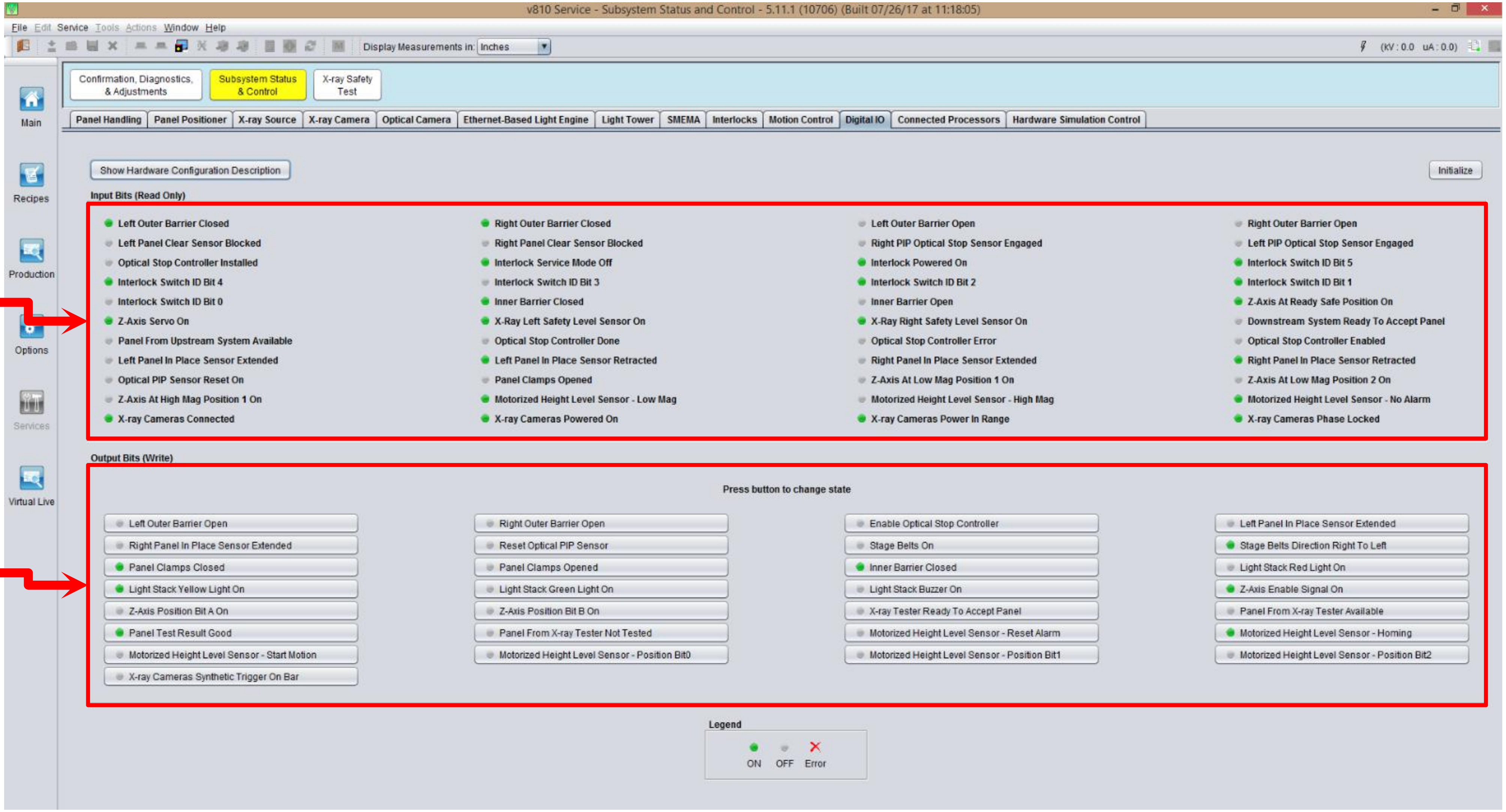

Input from Sensor as actual position feedback

Command provided to mechanism hardware

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GUI Interface - Panel Handling tab

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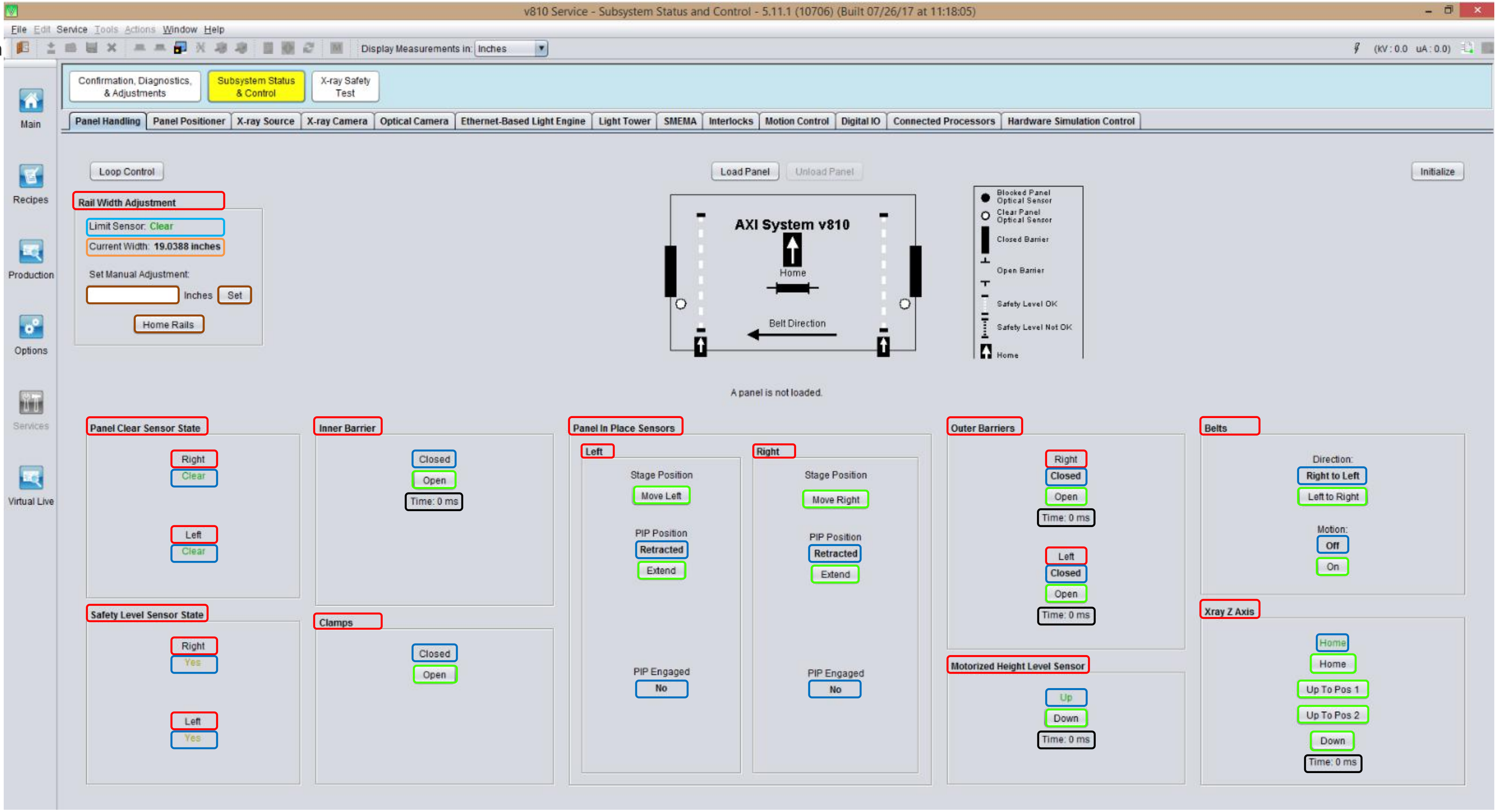

- Hardware
- Status Indicator (Digital Input Bit)
- Toggle (Digital Output Bit)
- Duration of hardware status change (Speed)
- AWA Home sensor status
- AWA Rail Width
- AWA Rail Width Setting

GUI Interface - Main Page - Manual Load Board

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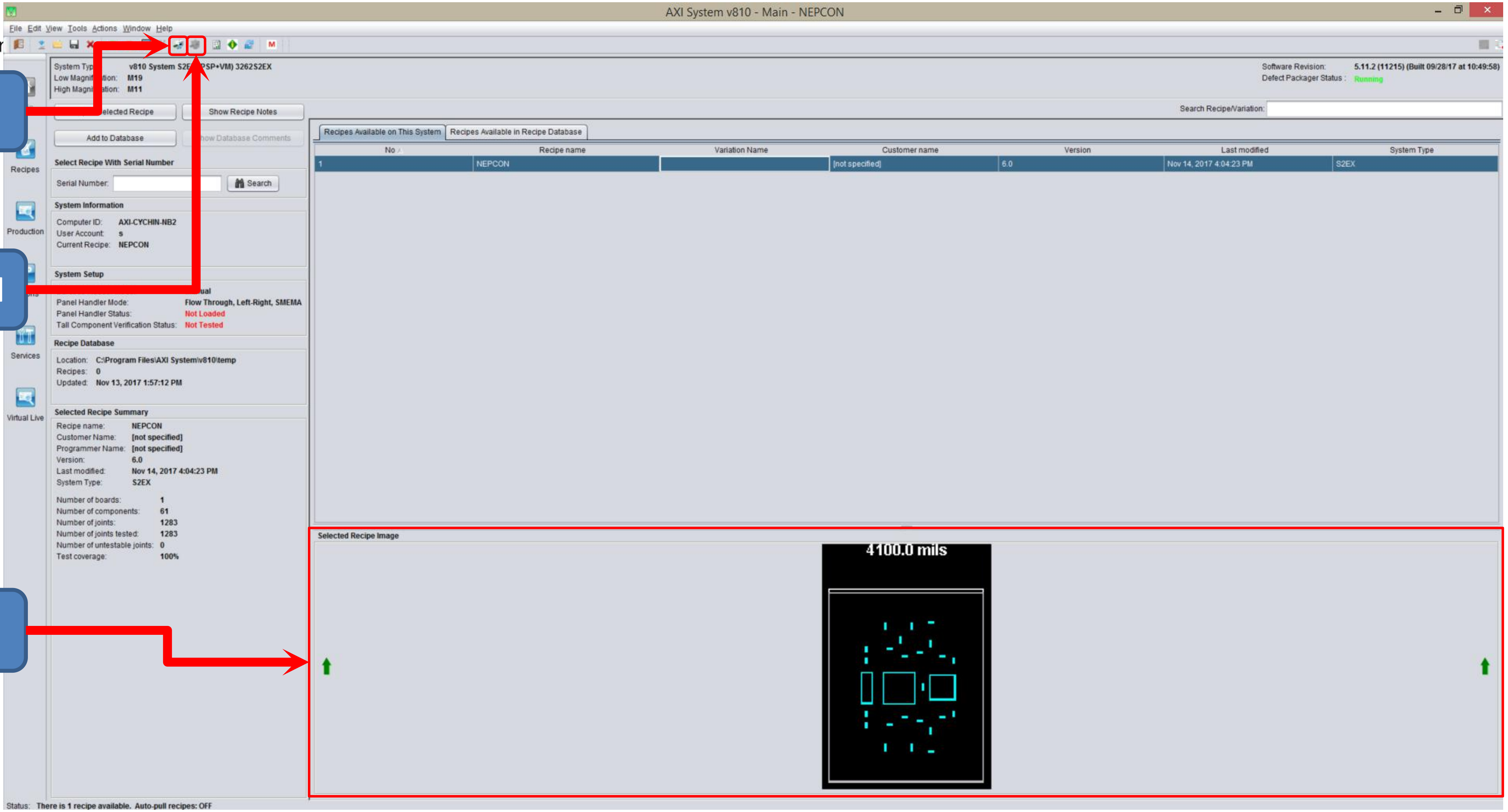

Only visible after select recipes

If loading direction change, image will change according

GUI Interface - Production Option - Panel Load Option

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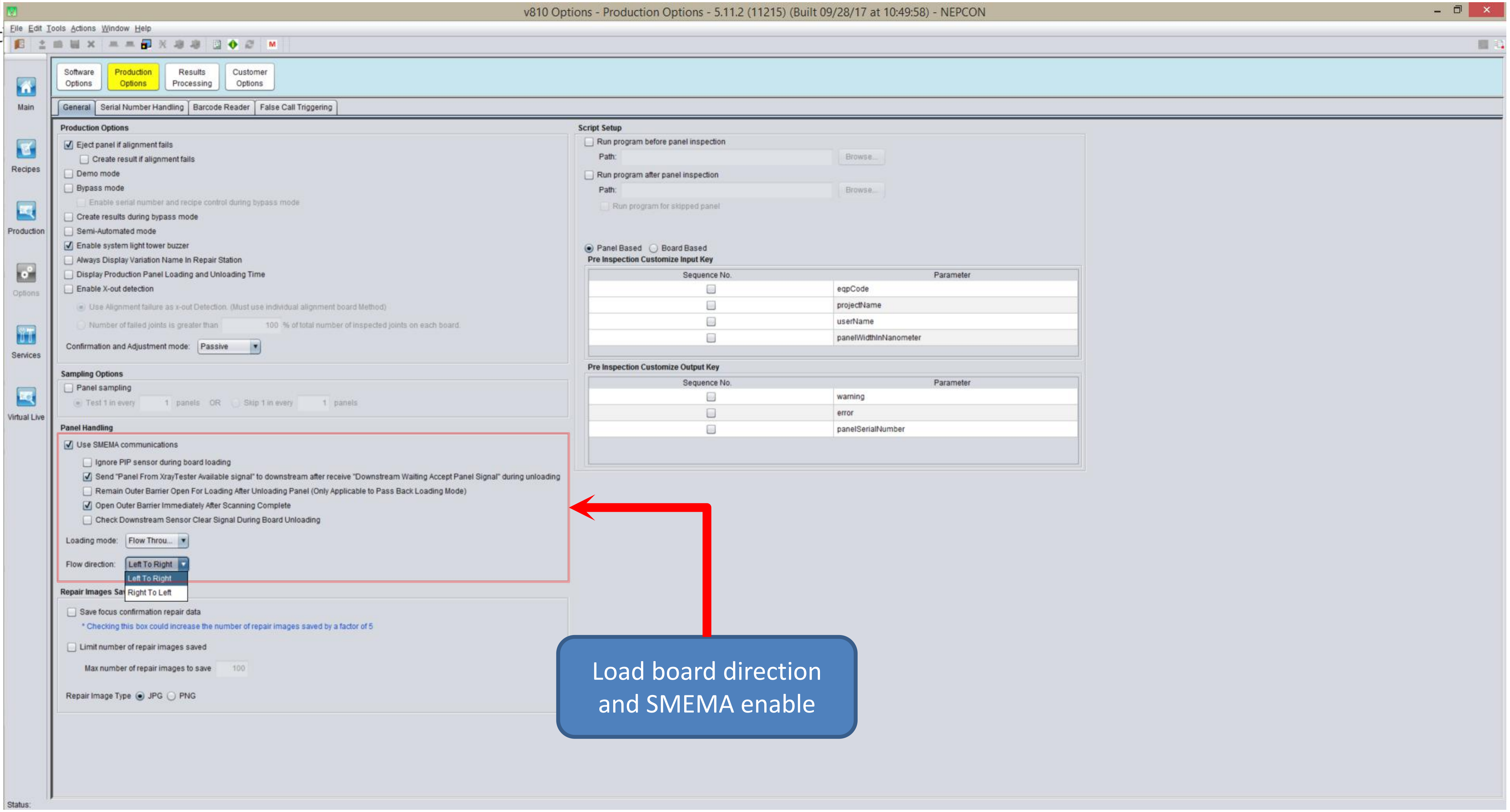

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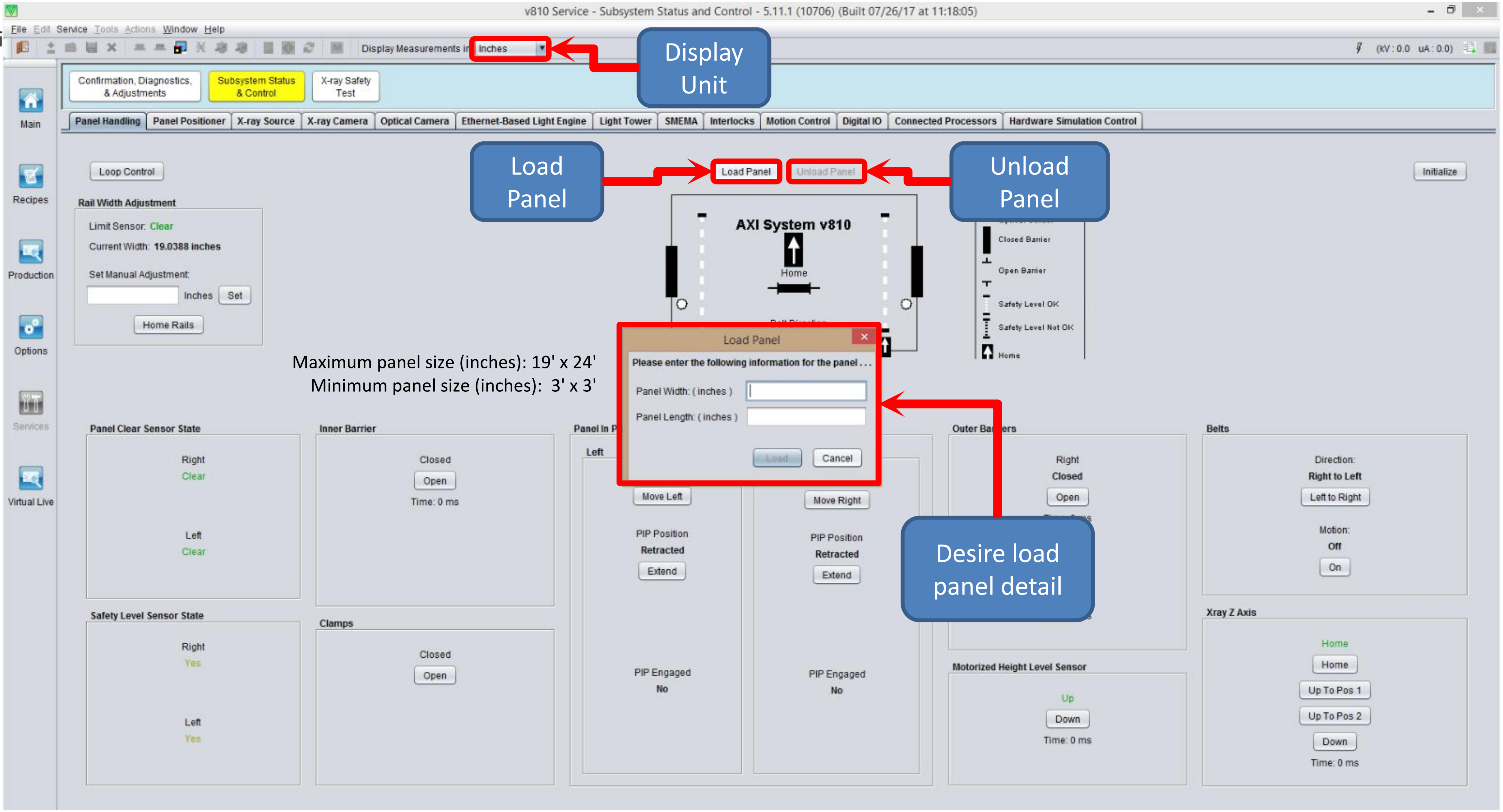
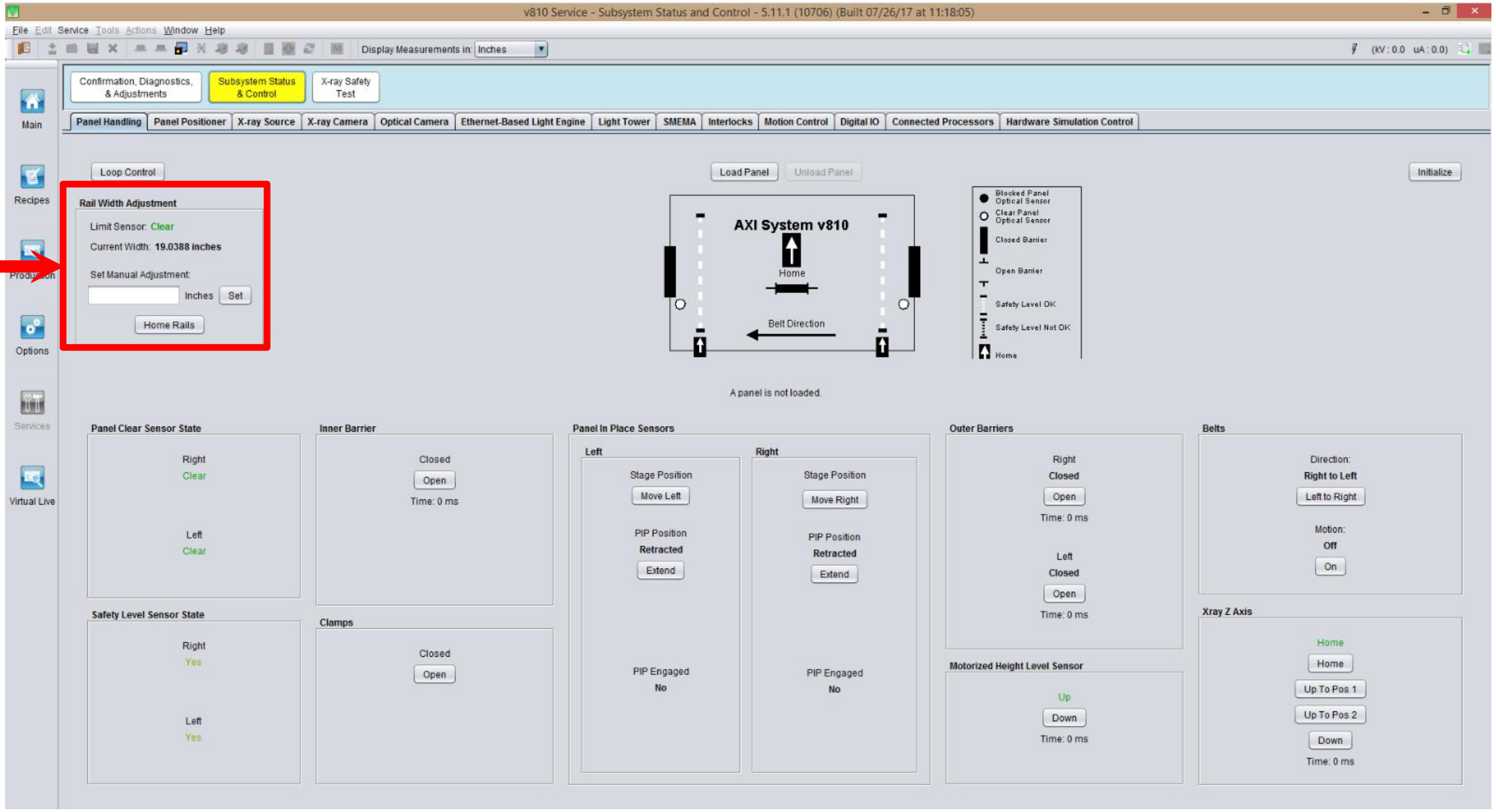
GUI Interface - Panel Handling Subsystem - Manual Load Board



Maximum panel size (inches): 19' x 24'
Minimum panel size (inches): 3' x 3'

GUI Interface - Panel Handling Subsystem - AWA Rail Width

AWA Rail Width self define



- AWA rail width home position and maximum rail width is 19 inches
- AWA minimum rail width is 3 inches
- AWA Limit sensor should always show “Clear”
- Manual set rail width value should same as physical measurement
- Panel loaded with toggle “Load Panel” will automatically + 0.3 mm as buffer to avoid incoming PCBA panel over width and board stuck at AWA

Hardware Overview - Panel Handling

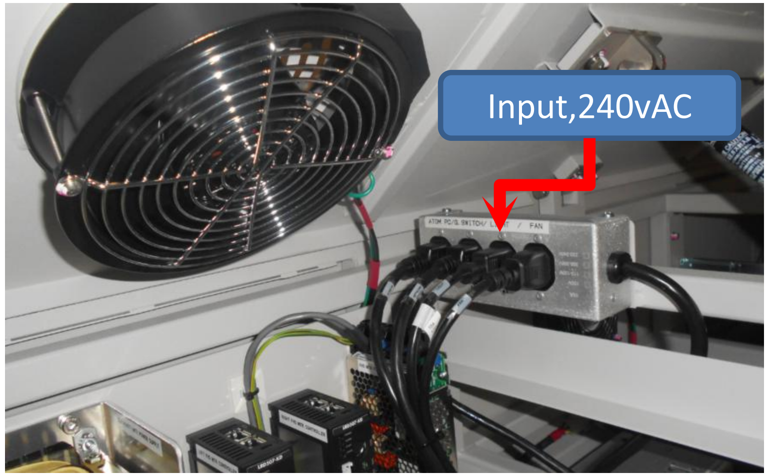
Automated Board Inspection

- Stage service lamp (Manual On/Off)
- Inner Barrier
- Outer Barriers
- XY Stage
  - Auto Width Adjustment (AWA)
  - Panel Clear Sensors
  - Belts, Motors, & Relays
  - Panel In Place Sensor (PIP)
  - Panel Clamps
- Motion Controller and Digital IO Assembly

Hardware Overview - Stage View Service Lamp

Left Service Lamp

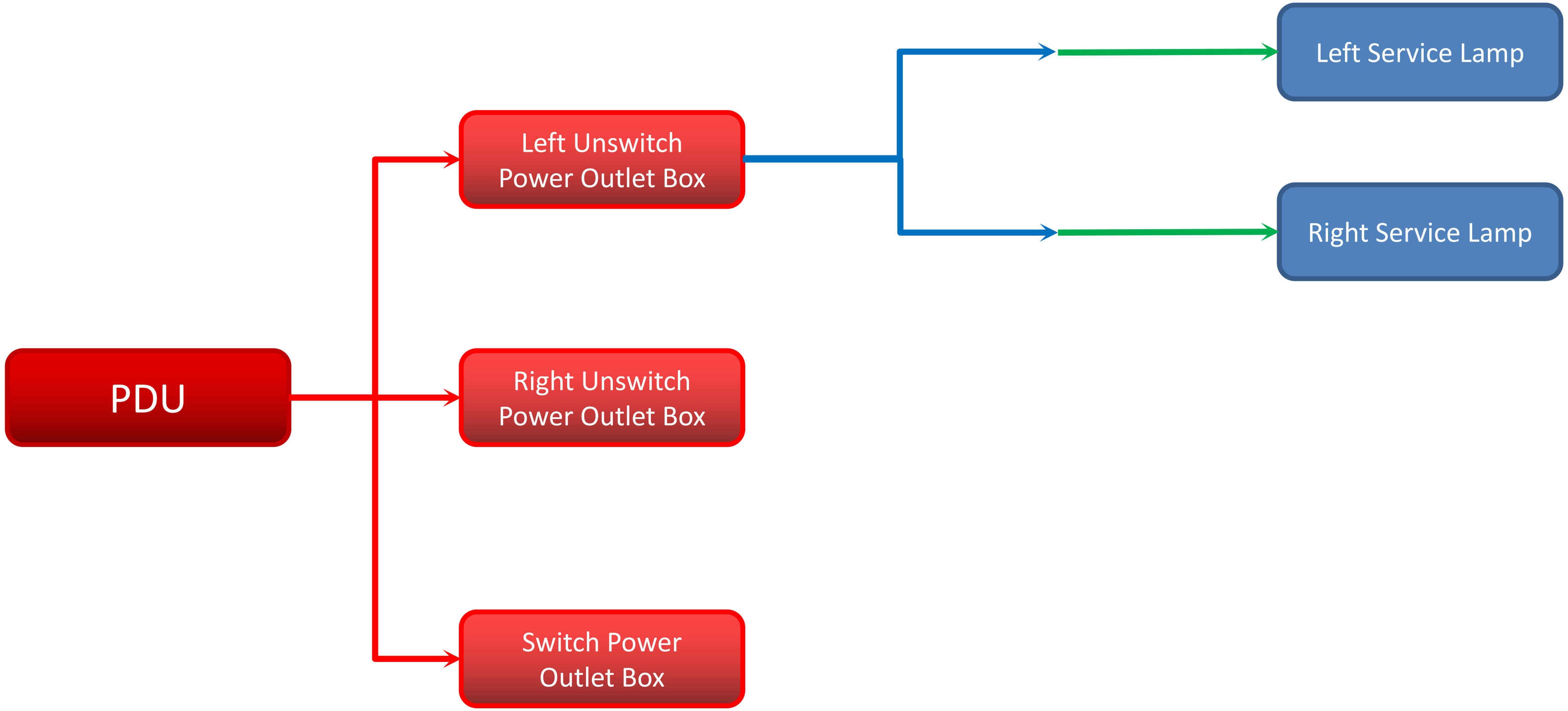
Right Service Lamp



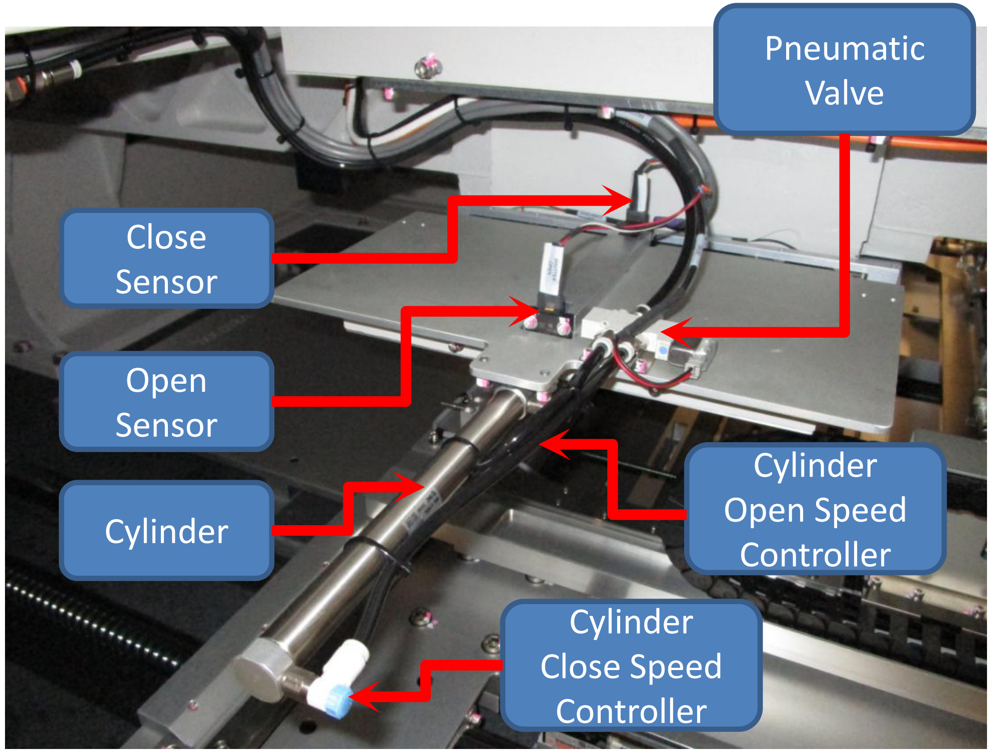
Left Server Rack, Unswitch Power Outlet Box

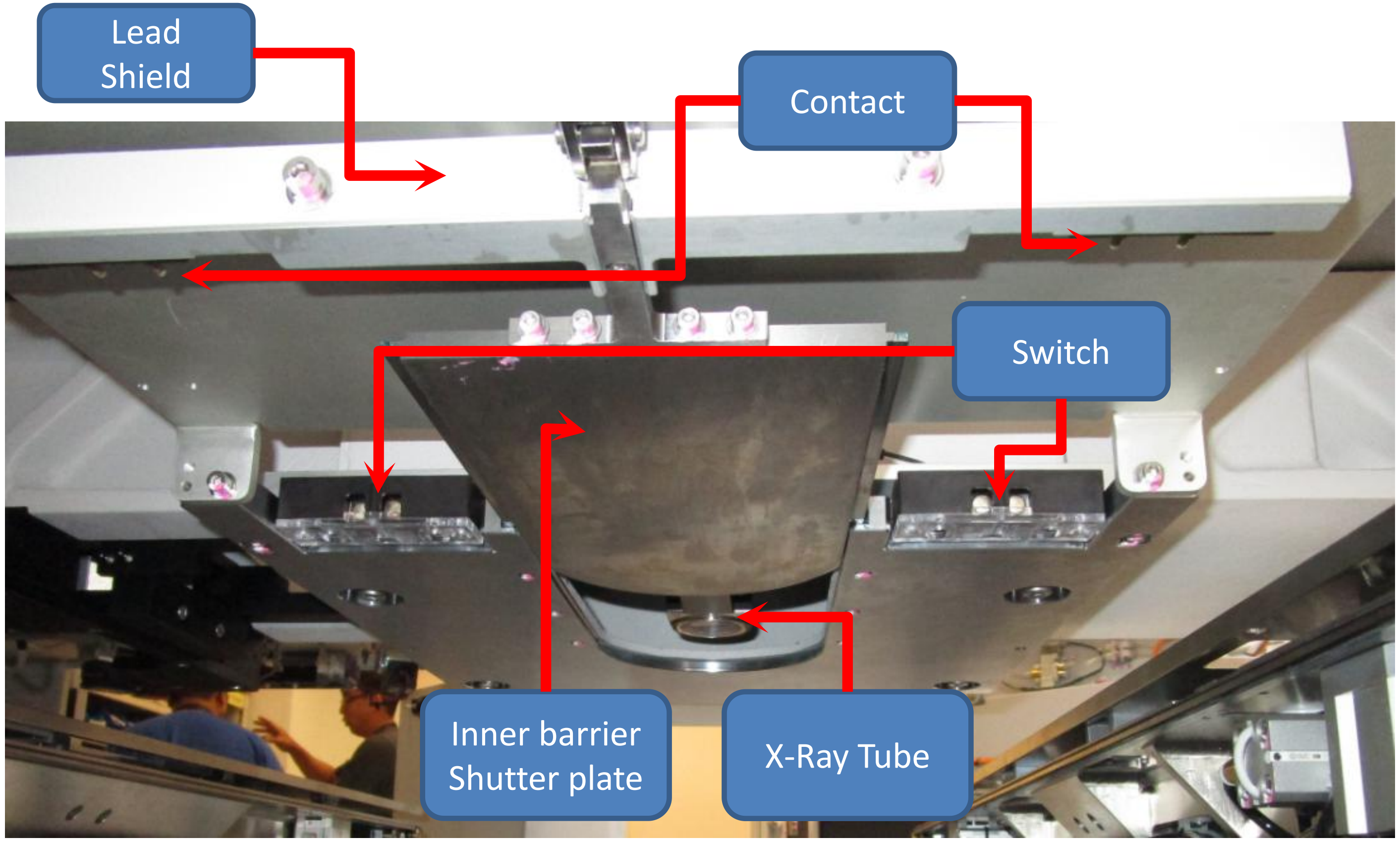
- Service lamp used is LED type, 9W, 570MM length, COOL WHITE
- Direct input from Unswitch Power Outlet Box, 240vAC
- Manual On/Off button

Hardware Overview - Stage View Service Lamp Connectivity



Hardware Overview - Inner Barrier





- ➢Opens to expose the panel to X-Rays for imaging.
- ➢Closes to contain X-Rays inside for loading and unloading the panel.

Hardware Overview - Interlock Connectivity

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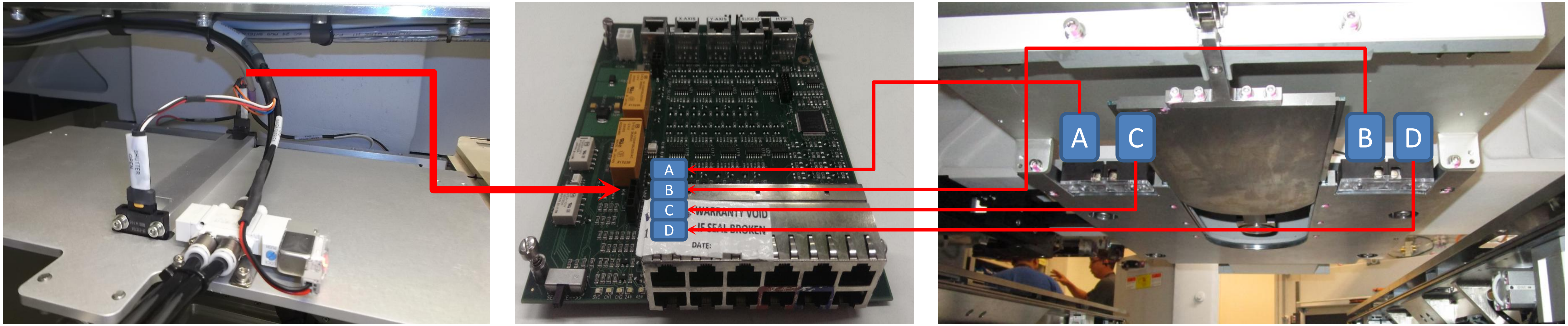

Hardware Overview - Interlock Connectivity Diagram

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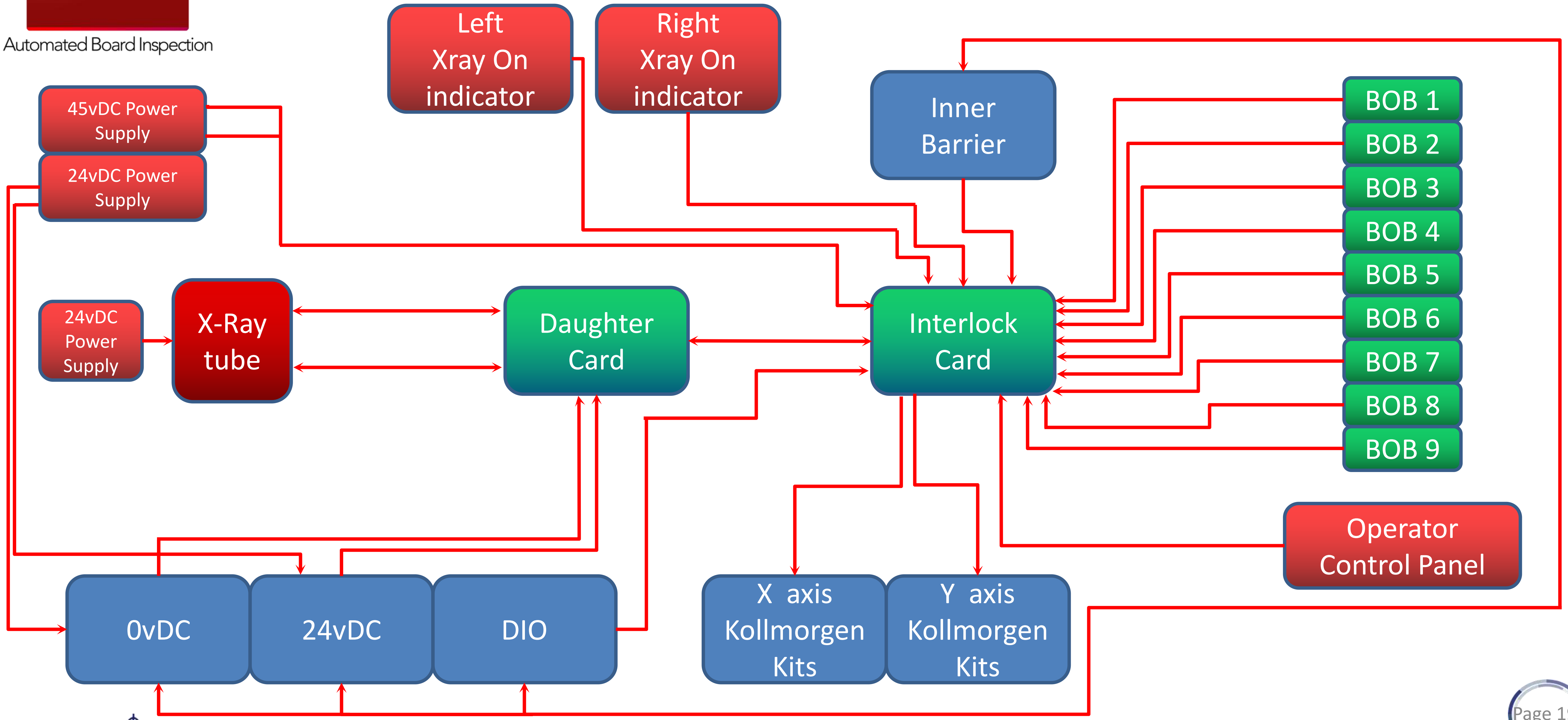
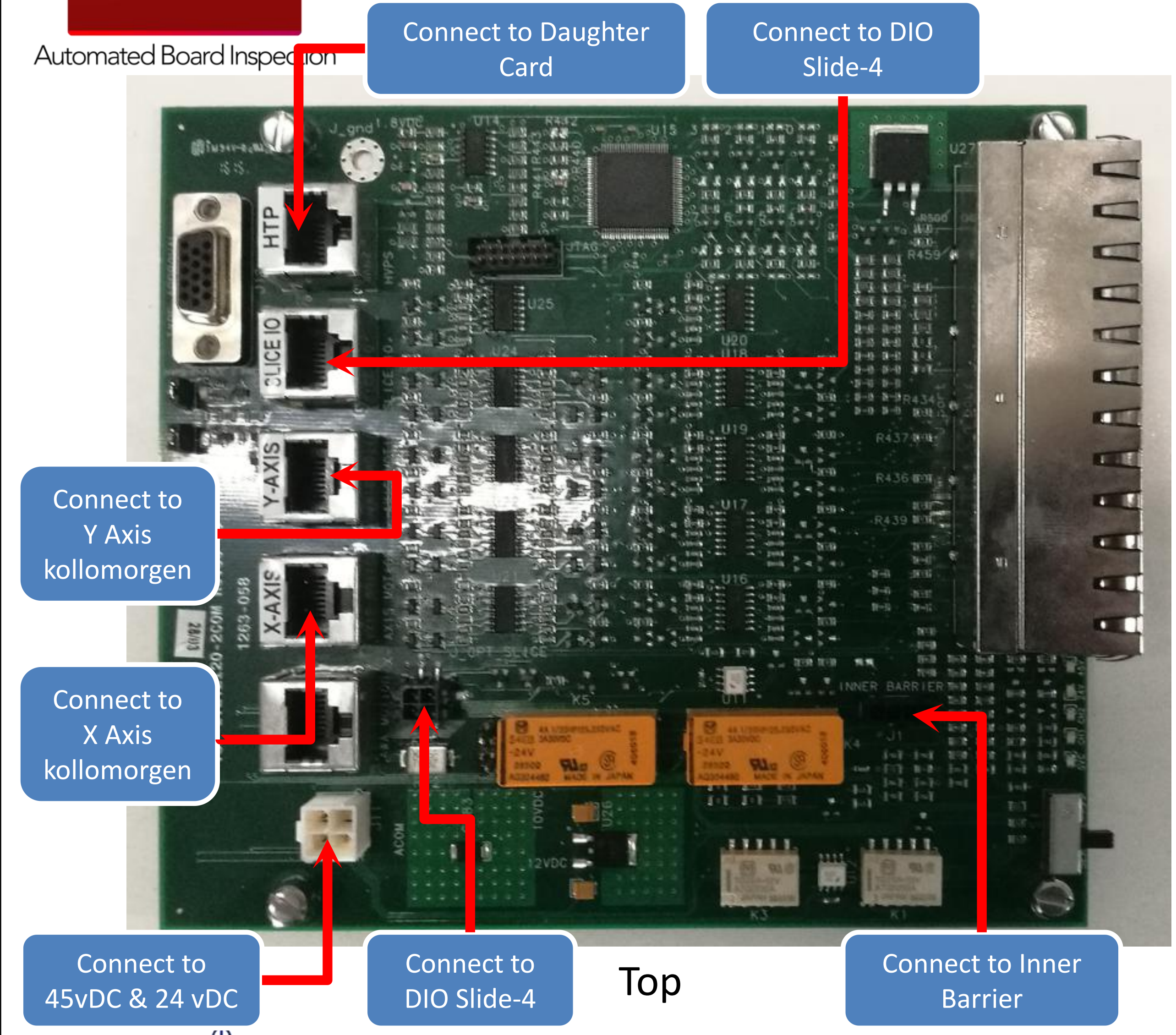
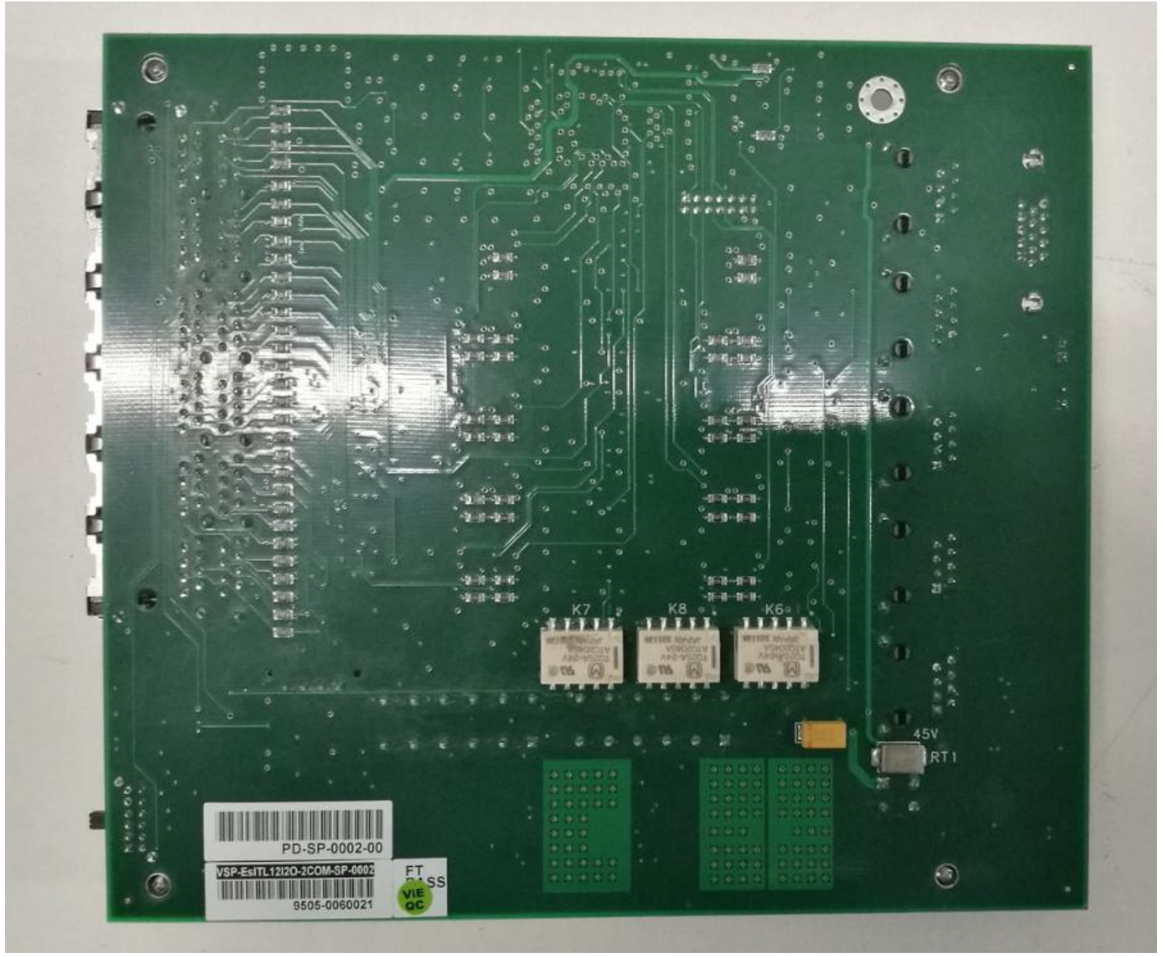


Hardware Overview - Inner Barrier - Interlock Card



Top



Bottom

Hardware Overview - Outer Barrier

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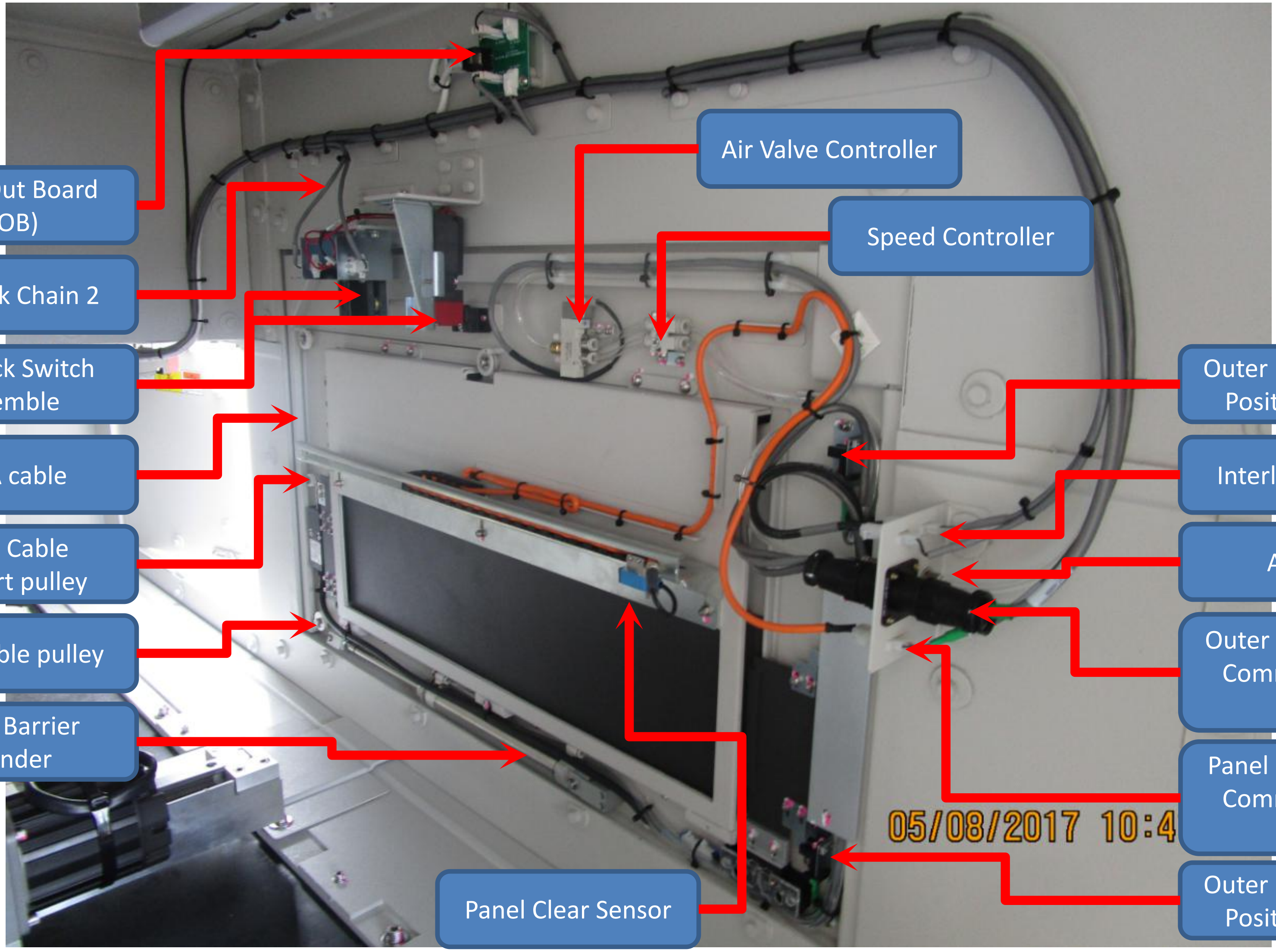
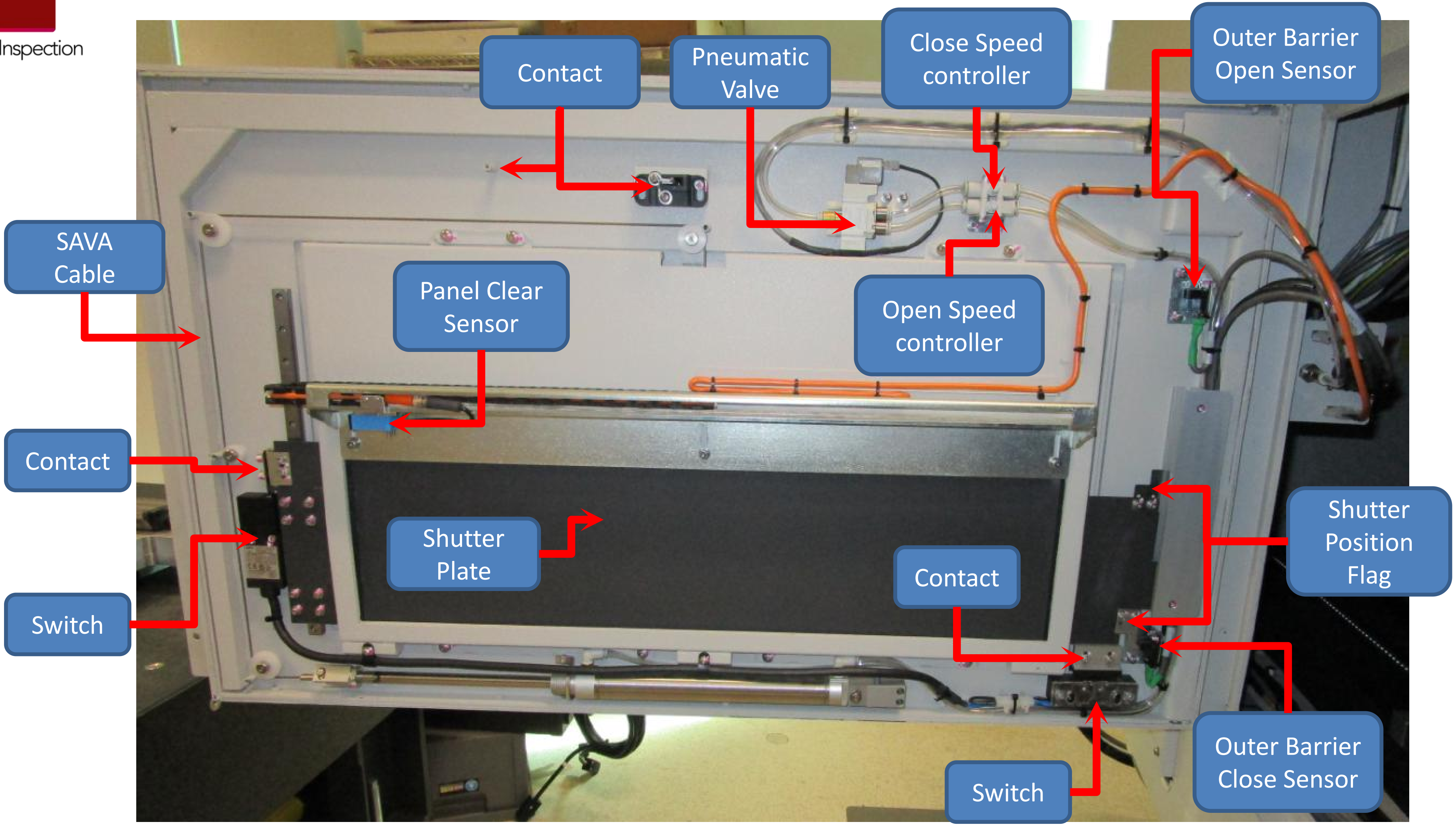
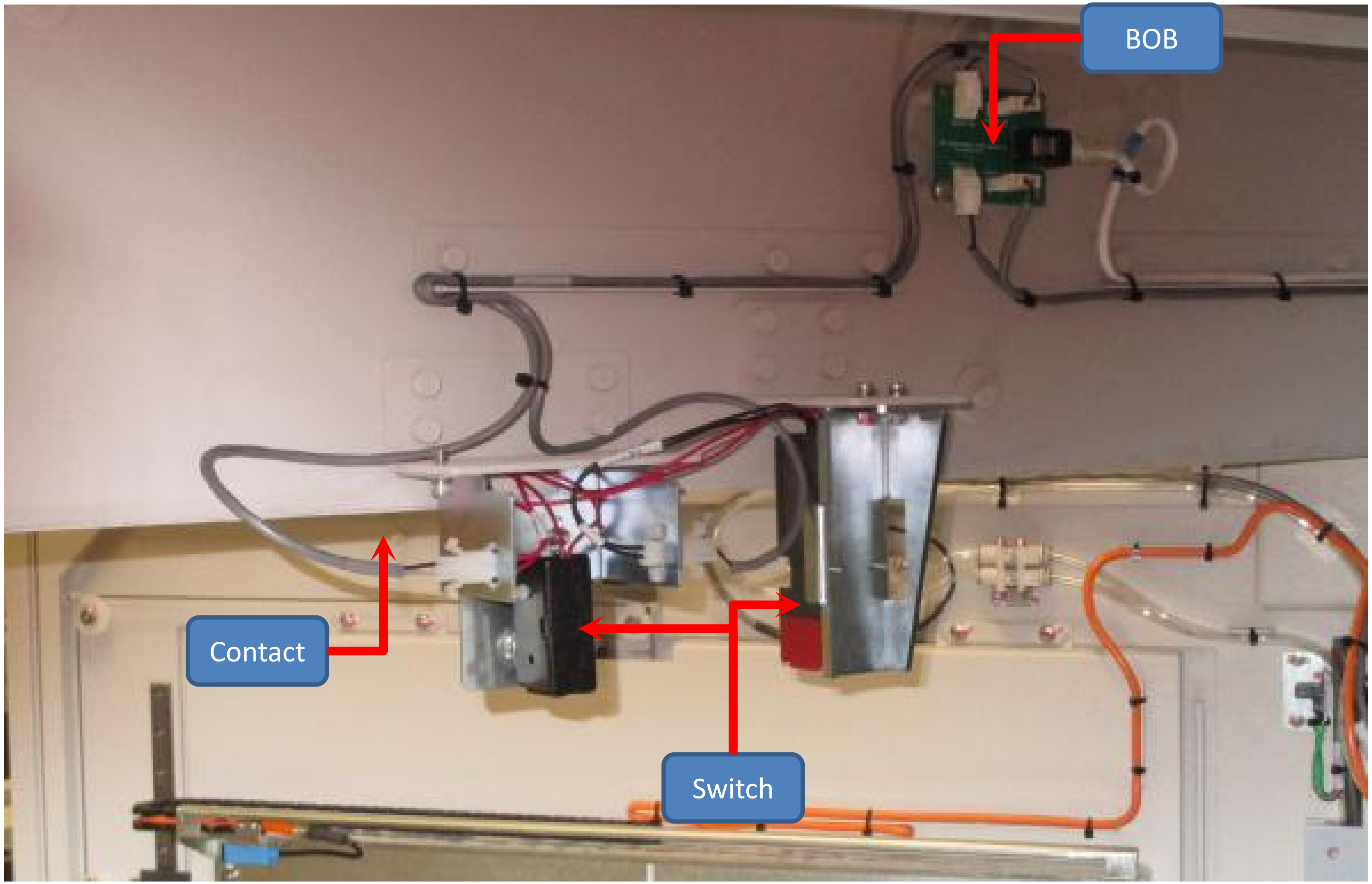


- Outer barrier open in order to load the board inside the machine

Hardware Overview - Outer Barrier



Hardware Overview - Outer Barrier



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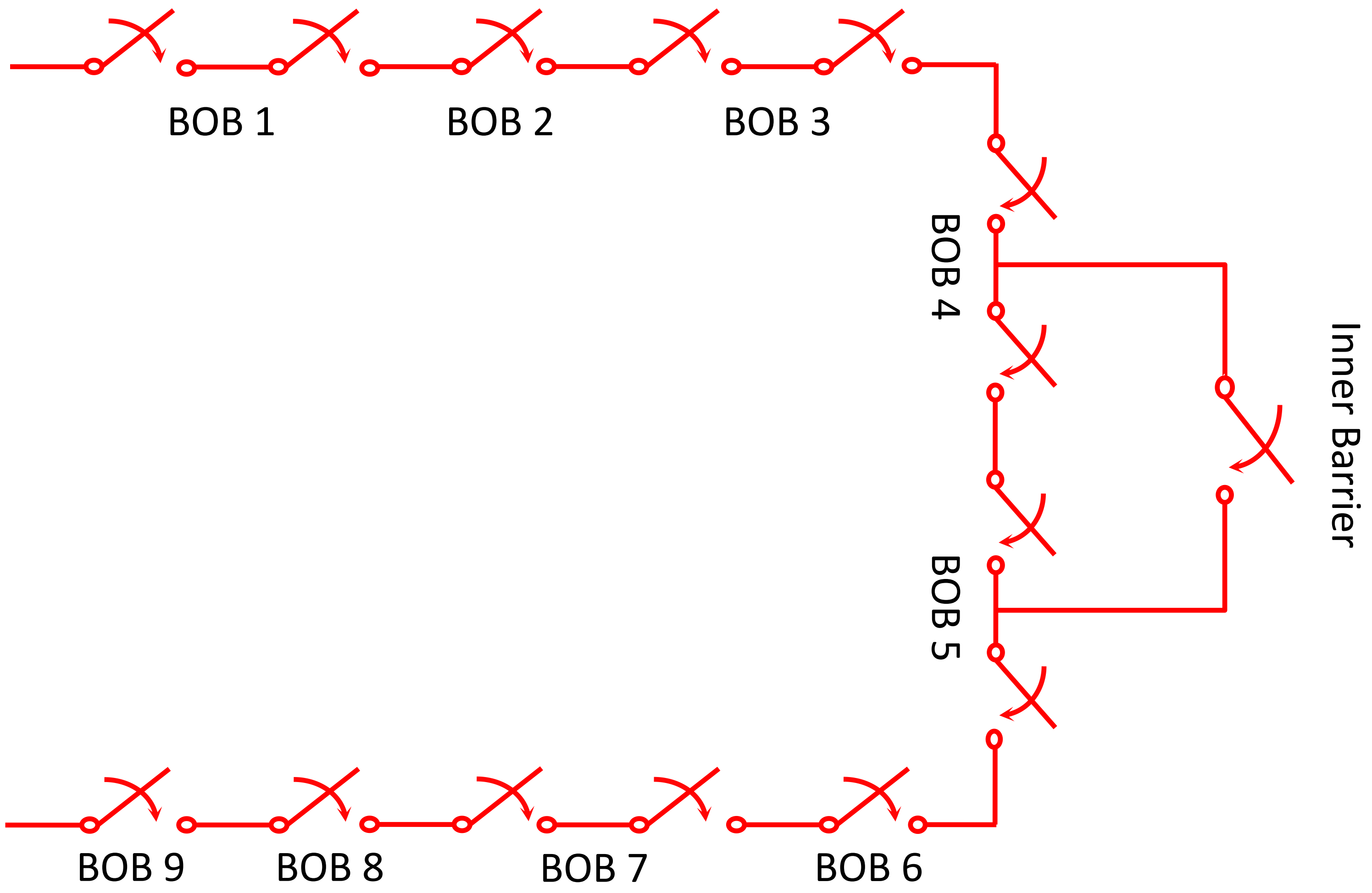
Interlock Subsystem Working Concept



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Hardware Overview - SSCA/MCA Box

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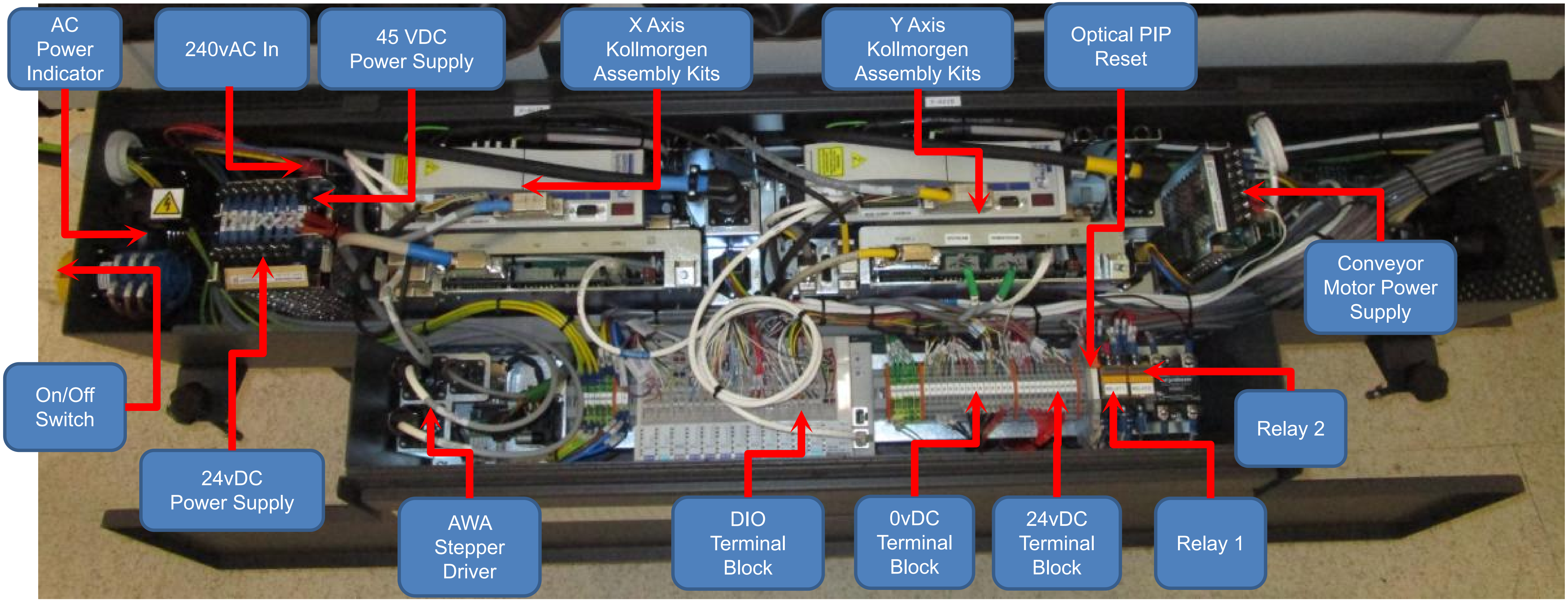

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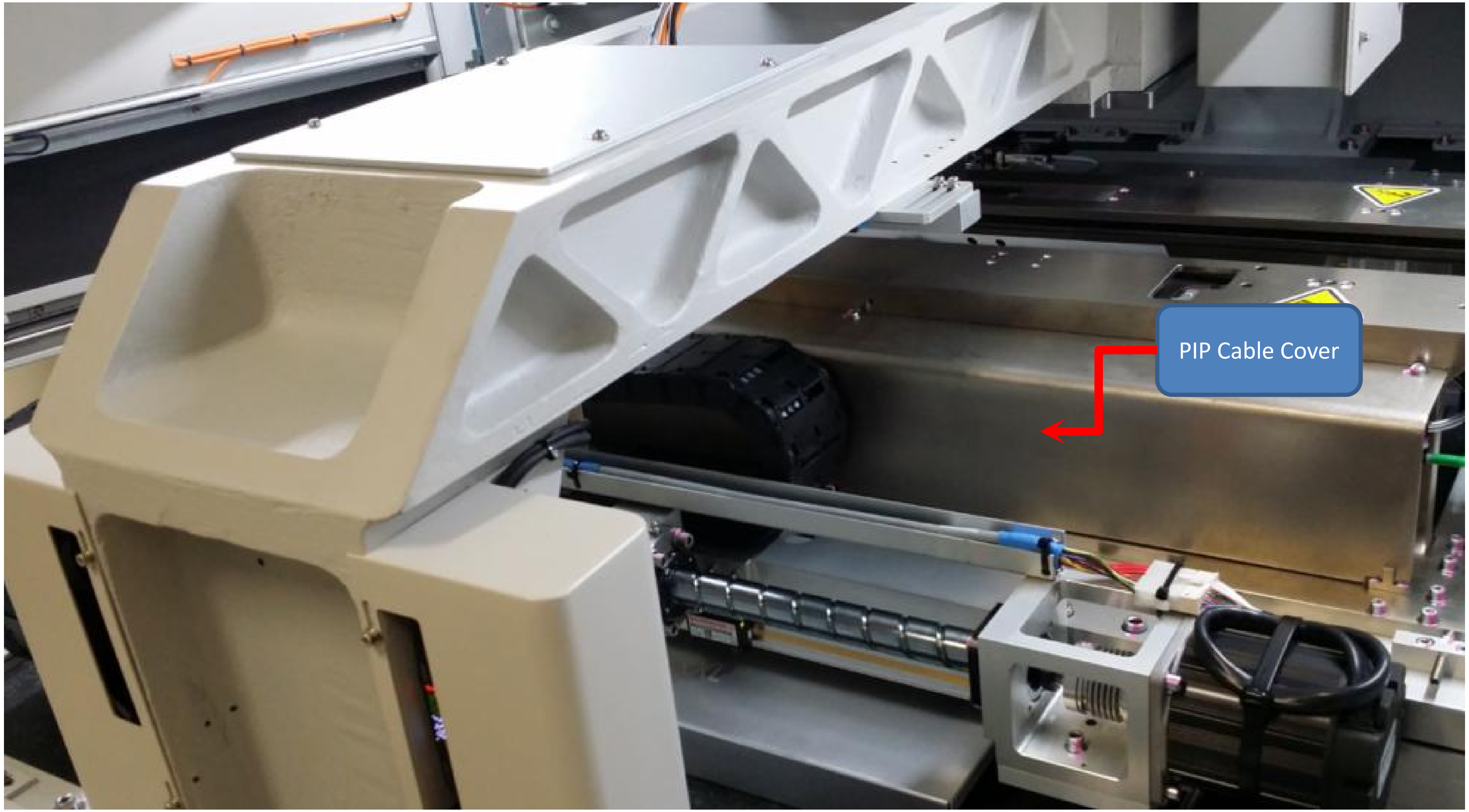
Hardware Overview - XY Stage



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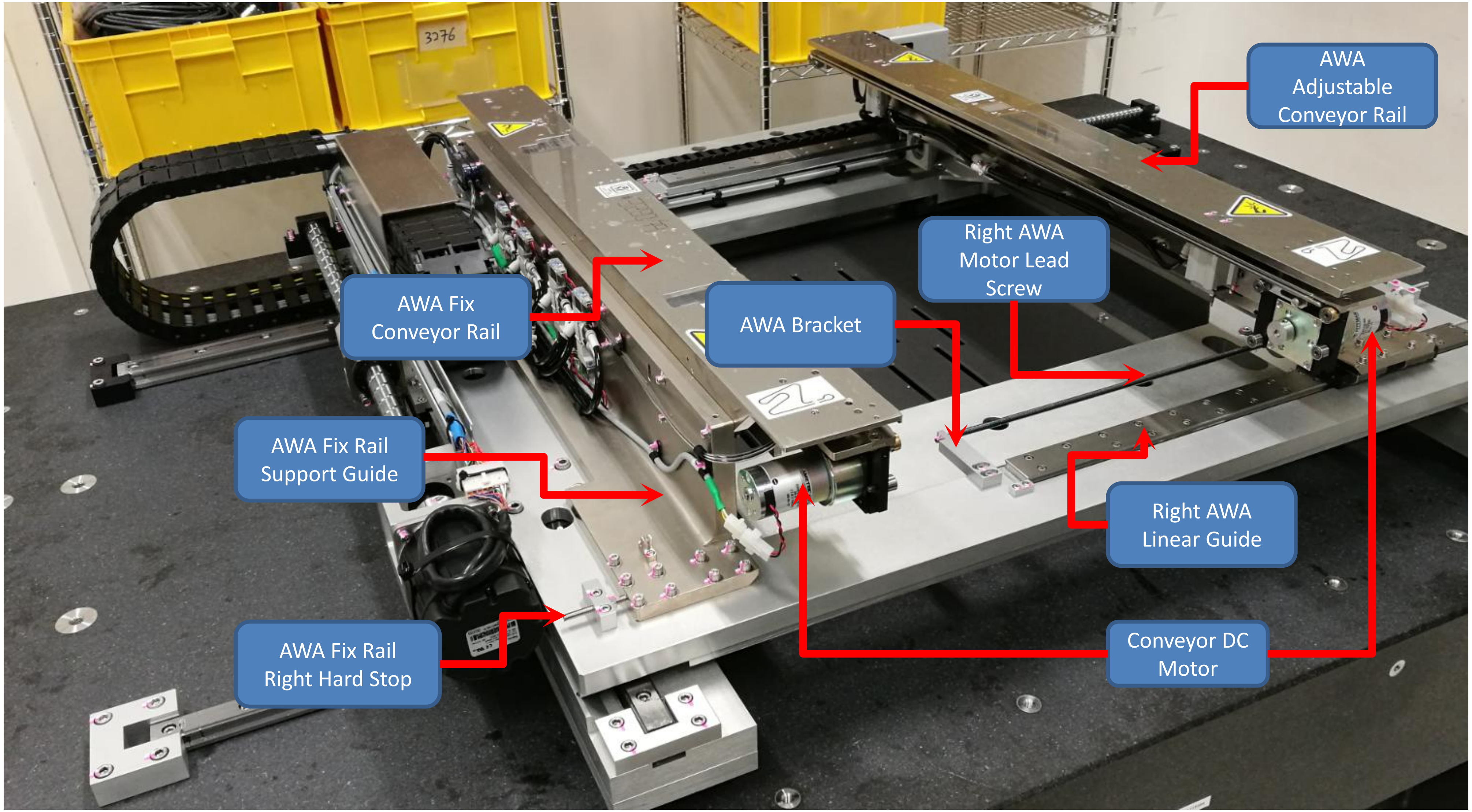
Hardware Overview - Panel Handling - XY Stage & AWA



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Hardware Overview - Panel Handling - XY Stage & AWA

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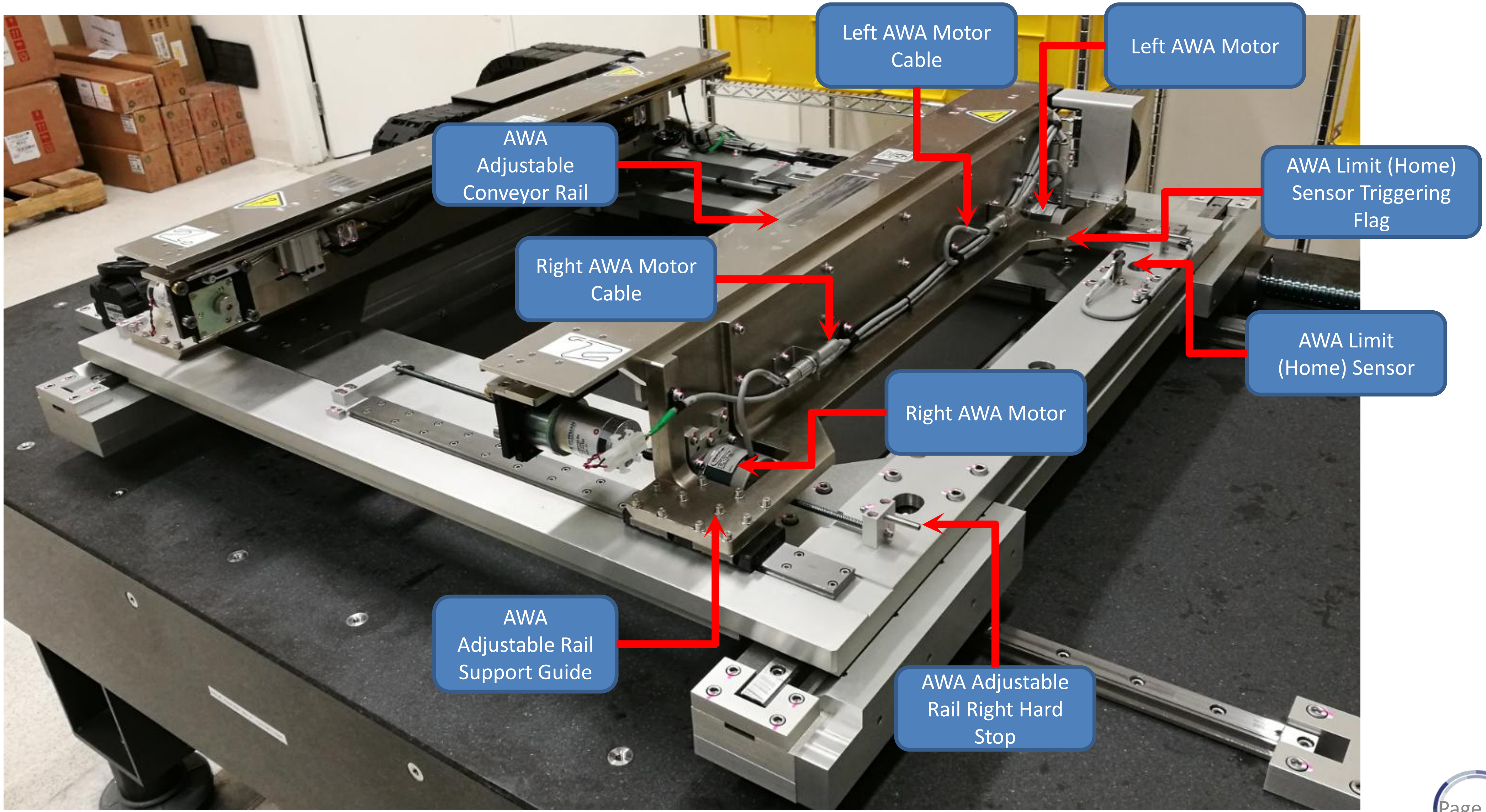

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Hardware Overview - Panel Handling - XY Stage & AWA

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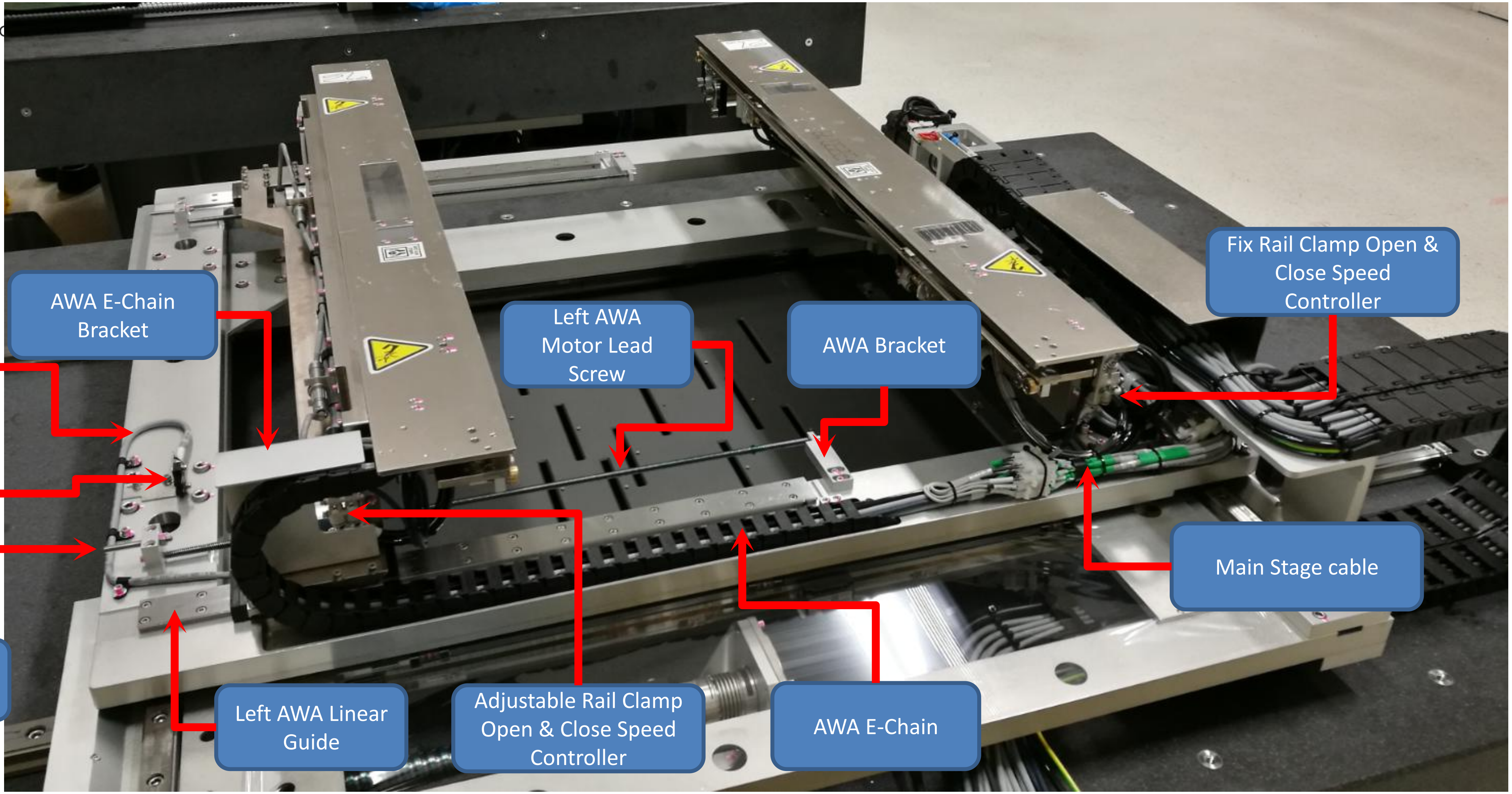

Hardware Overview - Panel Handling - XY Stage & AWA

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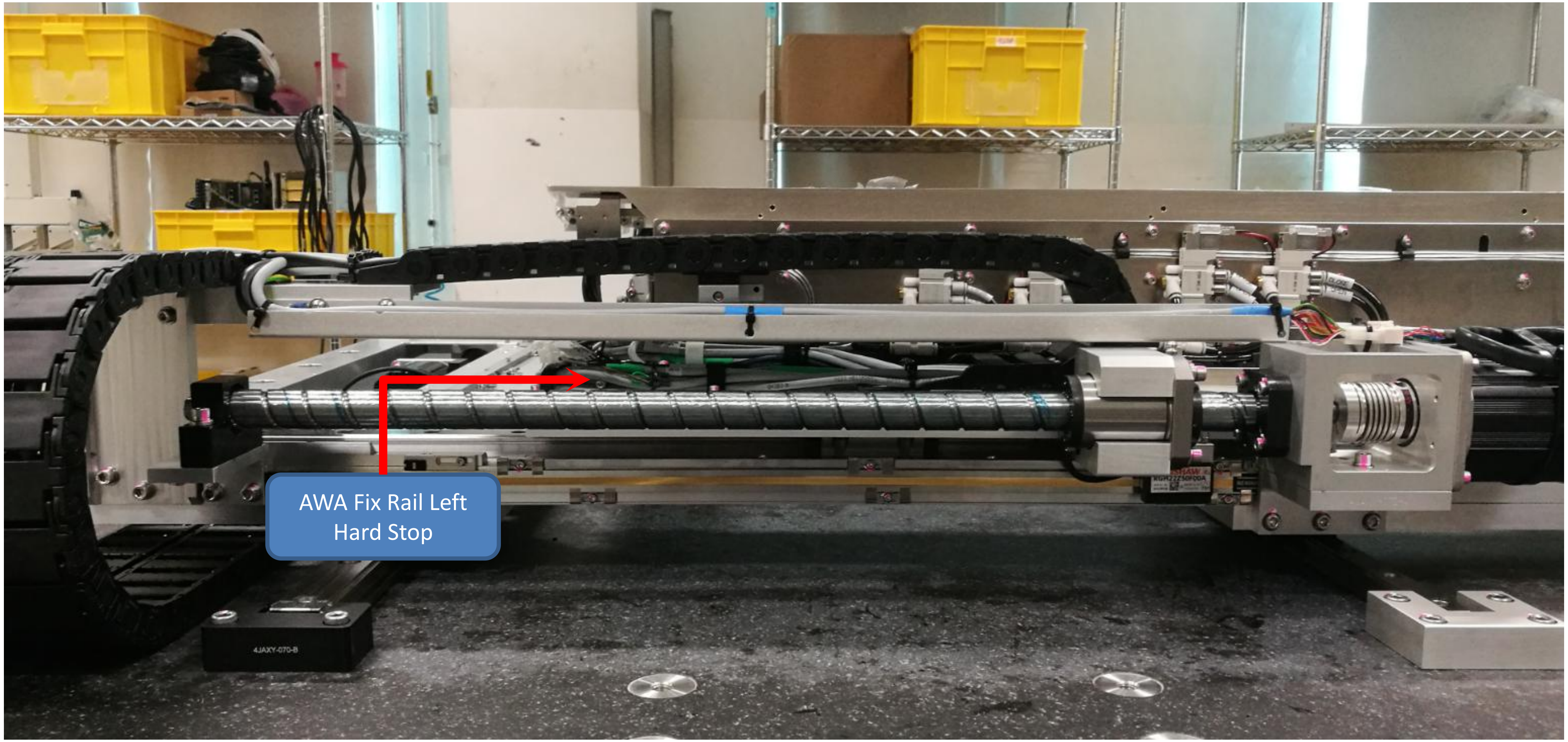

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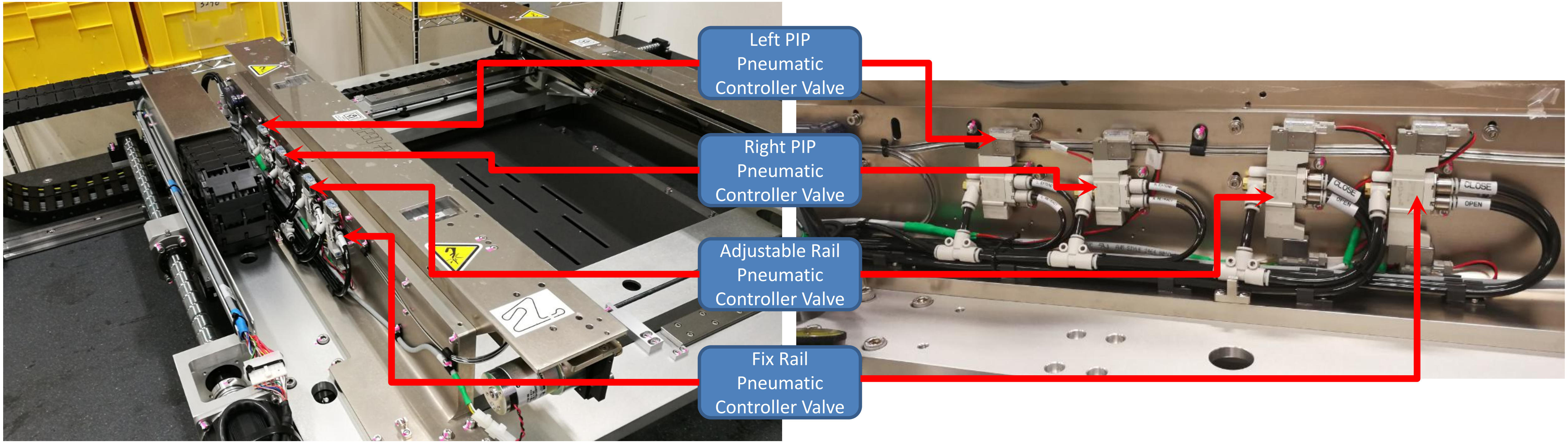
Hardware Overview - Panel Handling - Panel Clamp and PIP



Hardware Overview - Panel Handling - Panel Clamp

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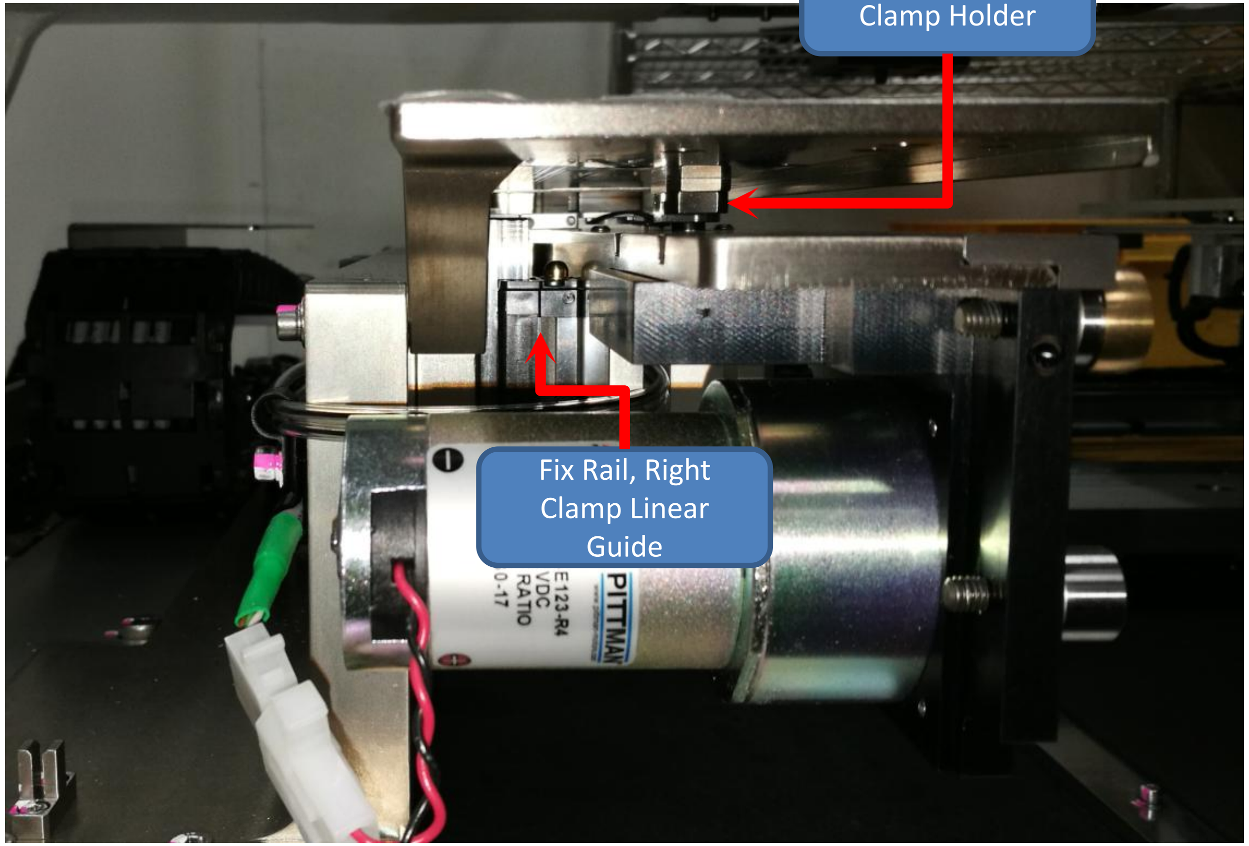
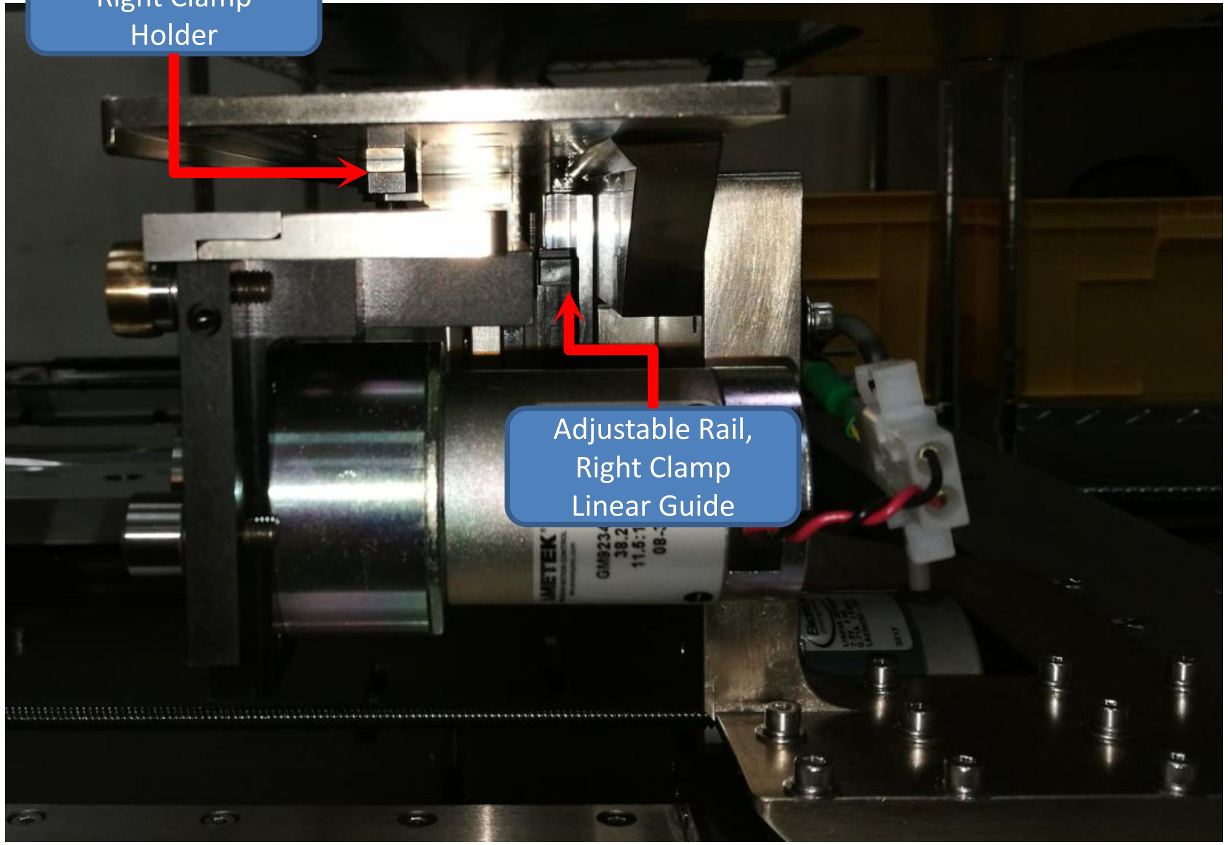
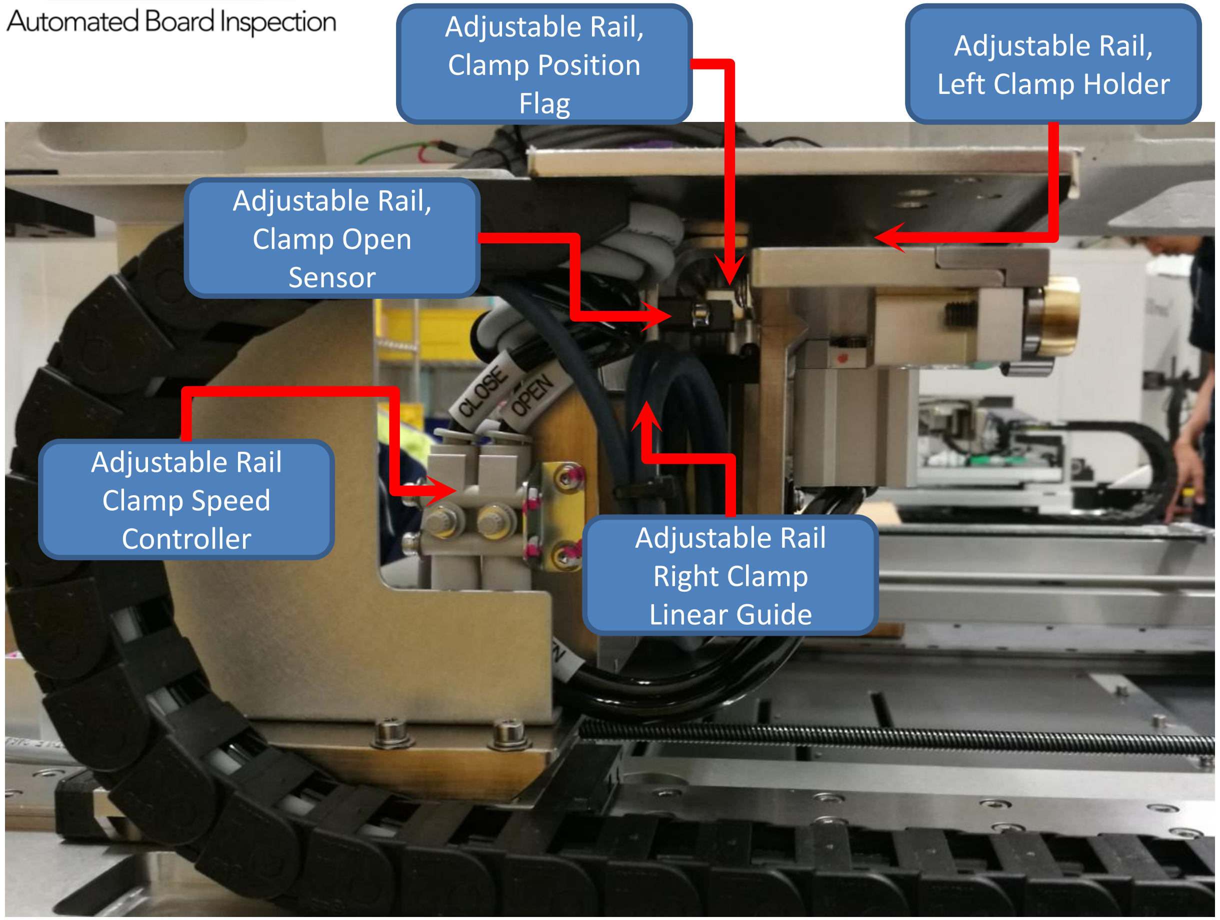
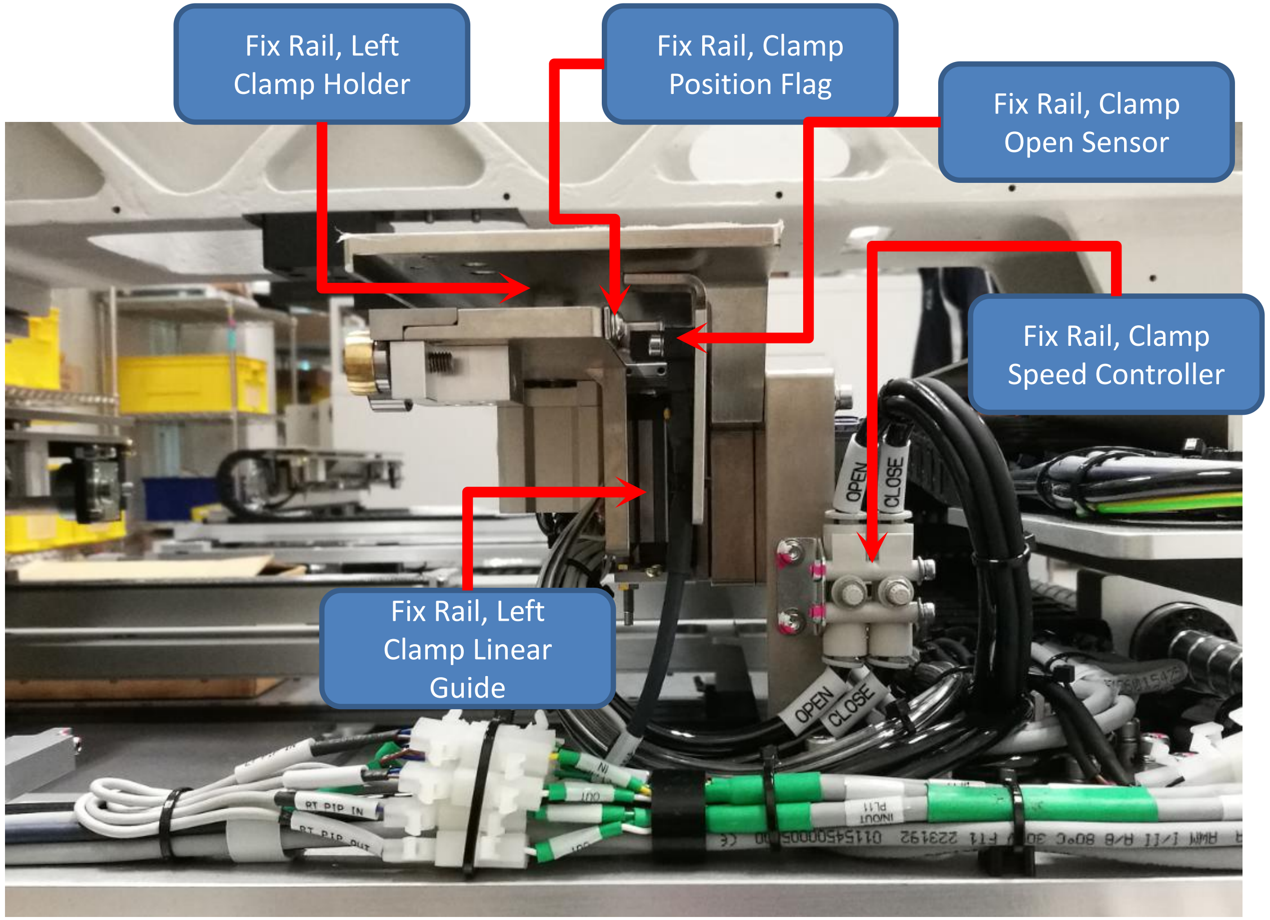




Hardware Overview - Panel Handling - Panel Clamp





Hardware Overview - XY Stage - Fix Rail

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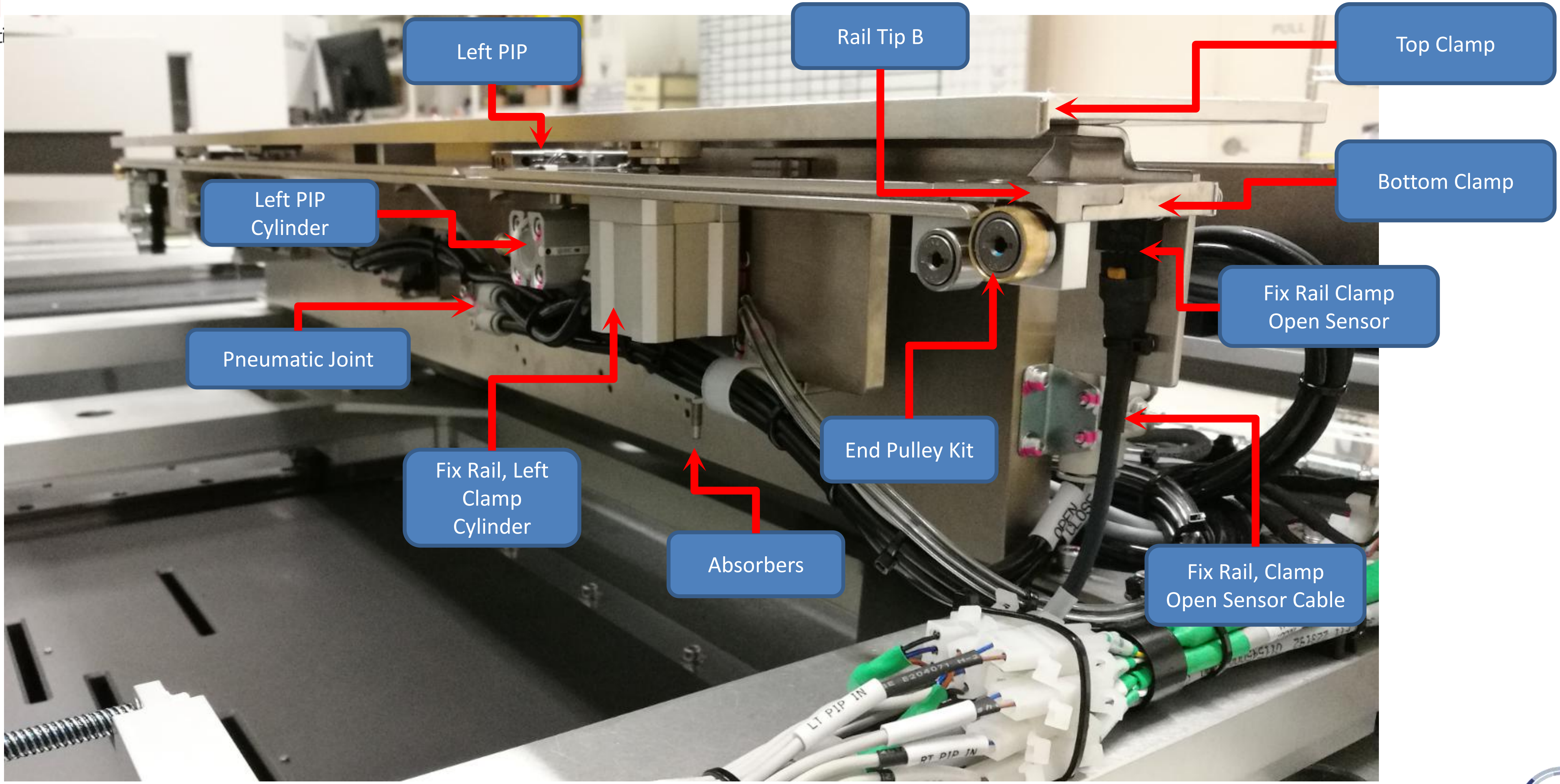

Hardware Overview - XY Stage - Fix Rail

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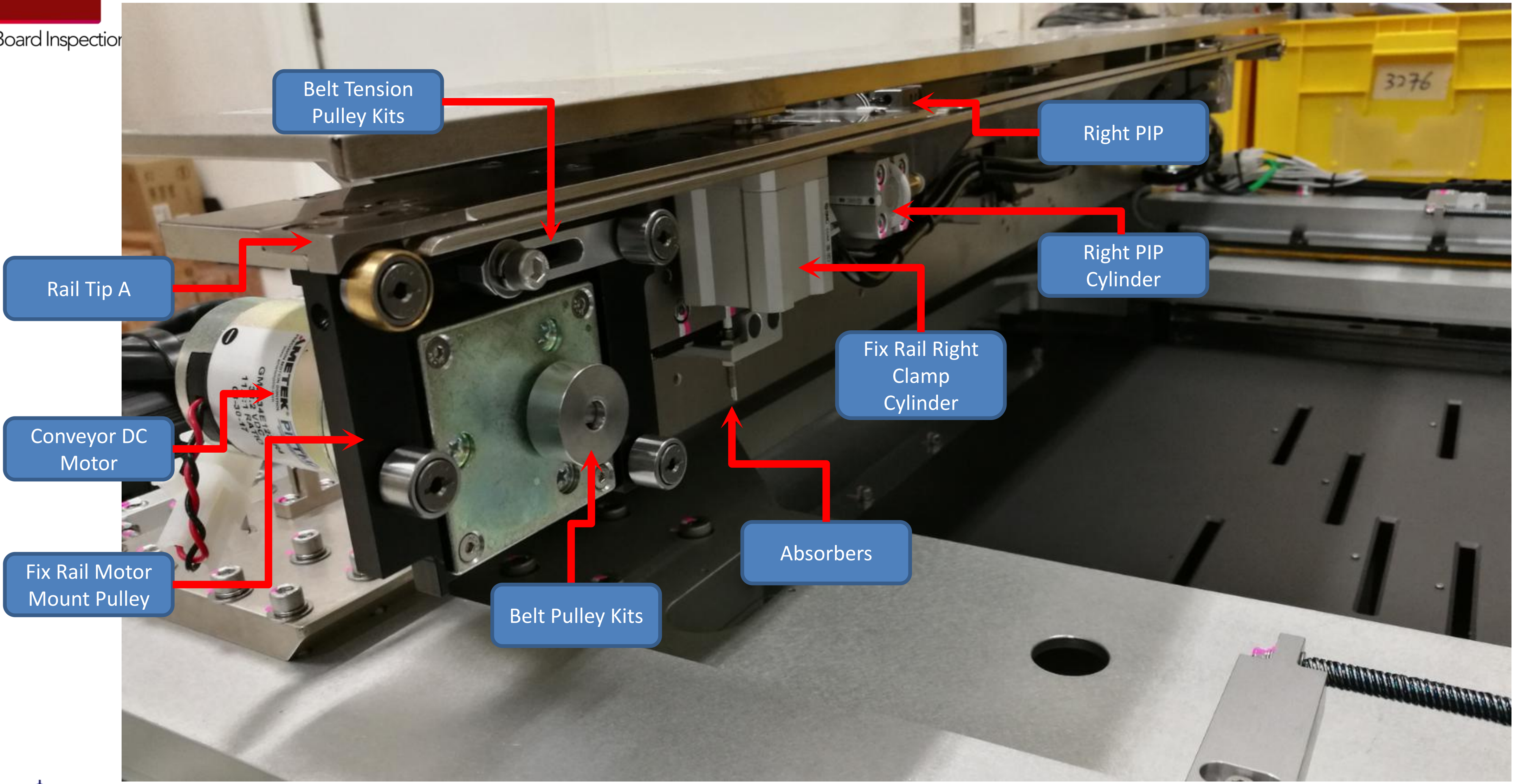

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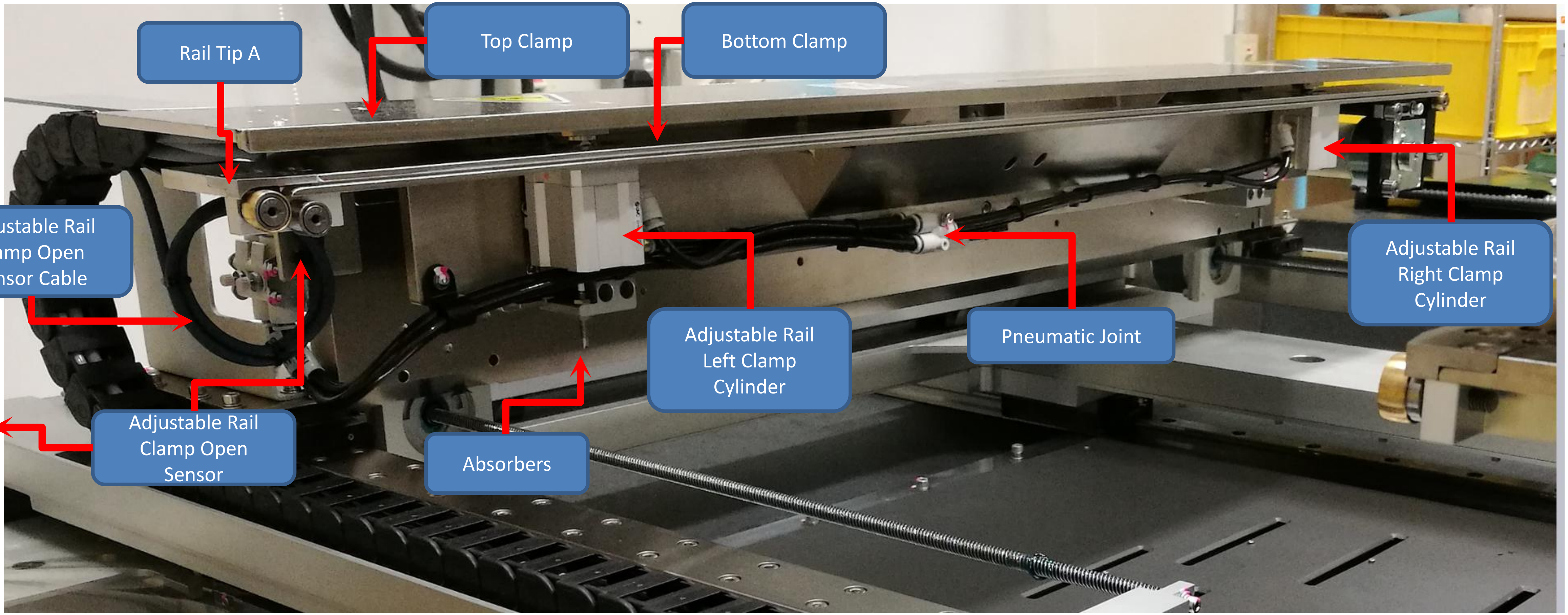
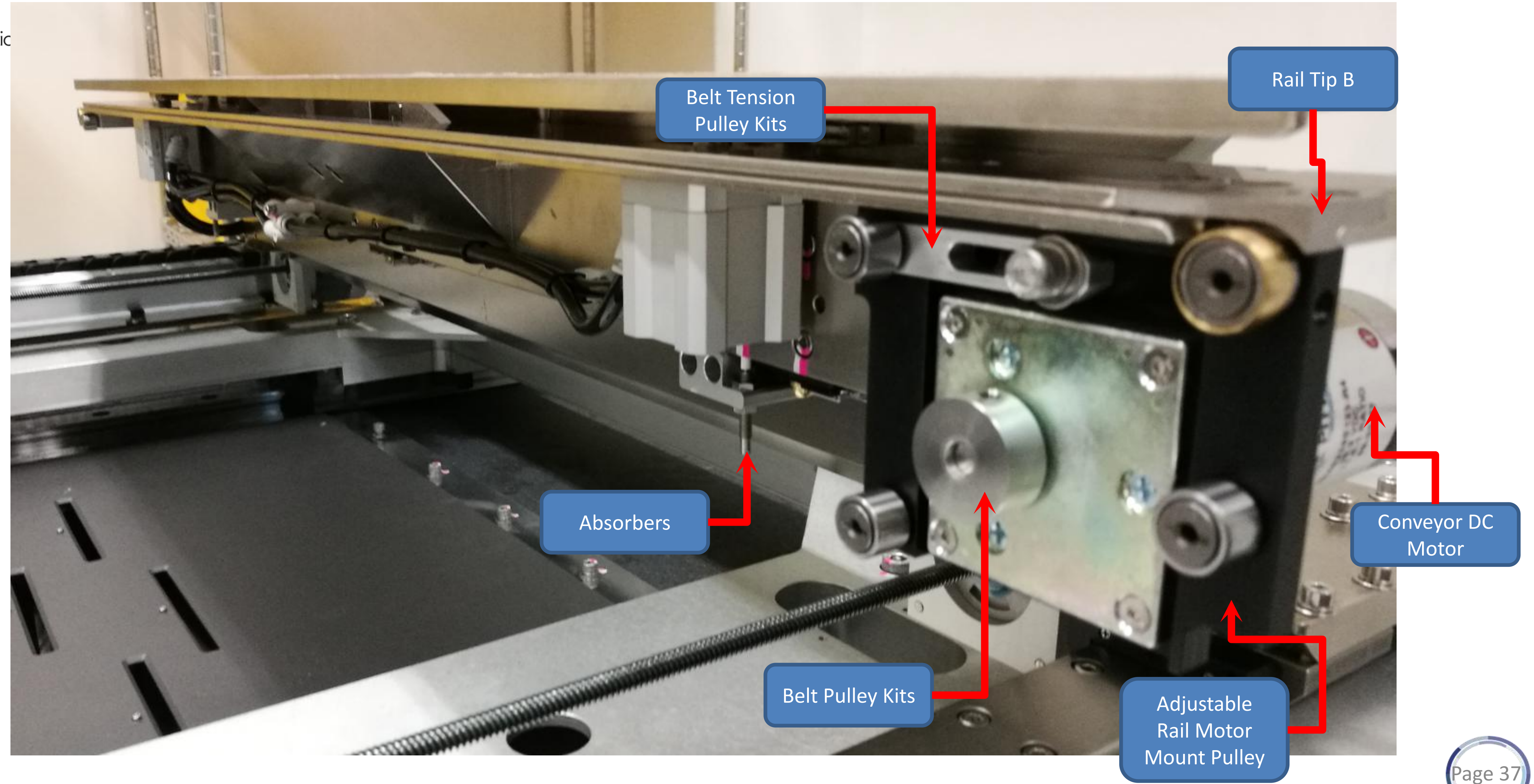
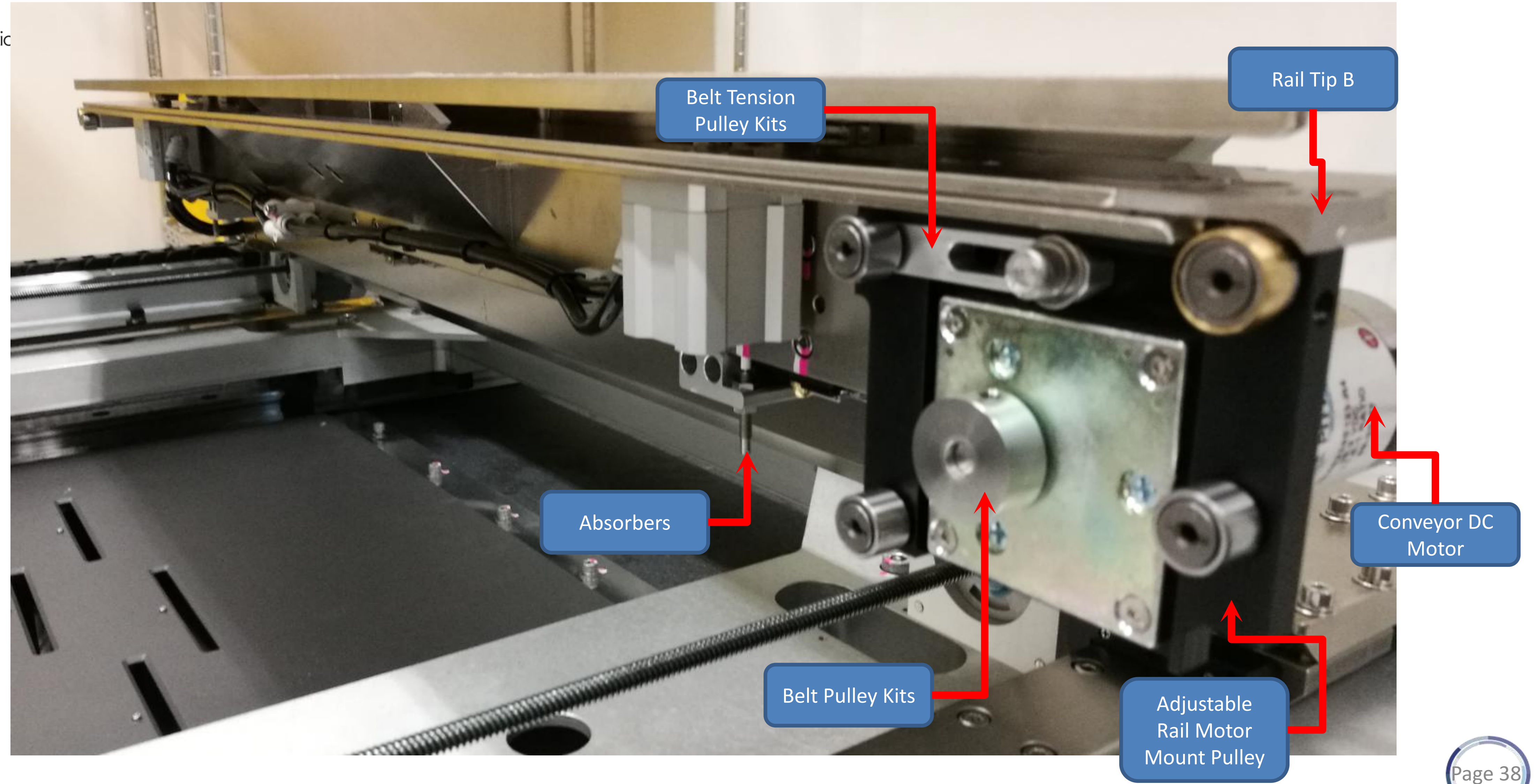
Hardware Overview - XY Stage - Adjustable Rail



Hardware Overview - XY Stage - Adjustable Rail



Hardware Overview - XY Stage - Adjustable Rail



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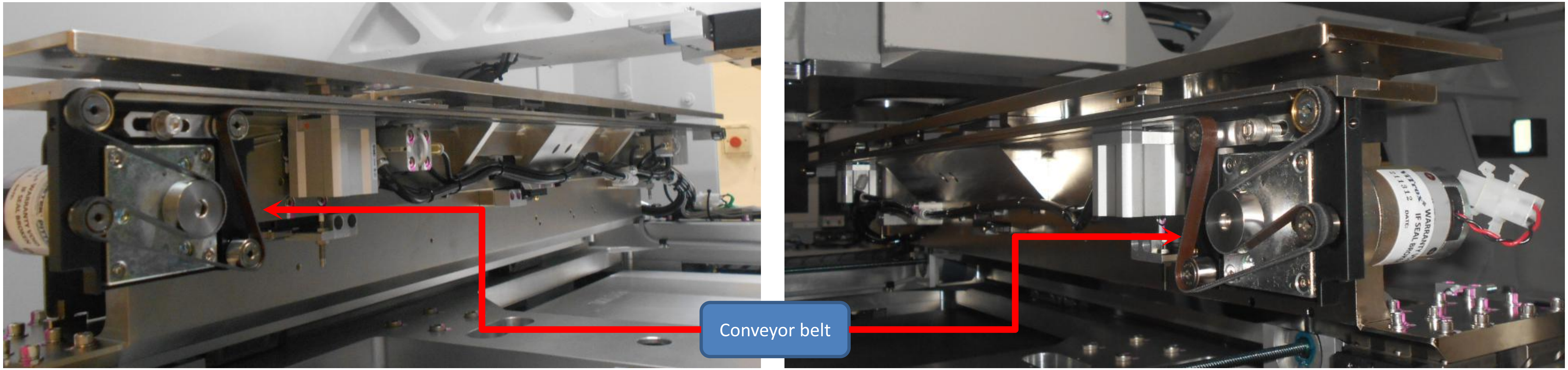
Hardware Overview - XY Stage - Conveyor Belt



Fix Rail Conveyor Belt Routing

Adjustable Rail Conveyor Belt Routing

Hardware Overview - Auto Width Adjustable (AWA)

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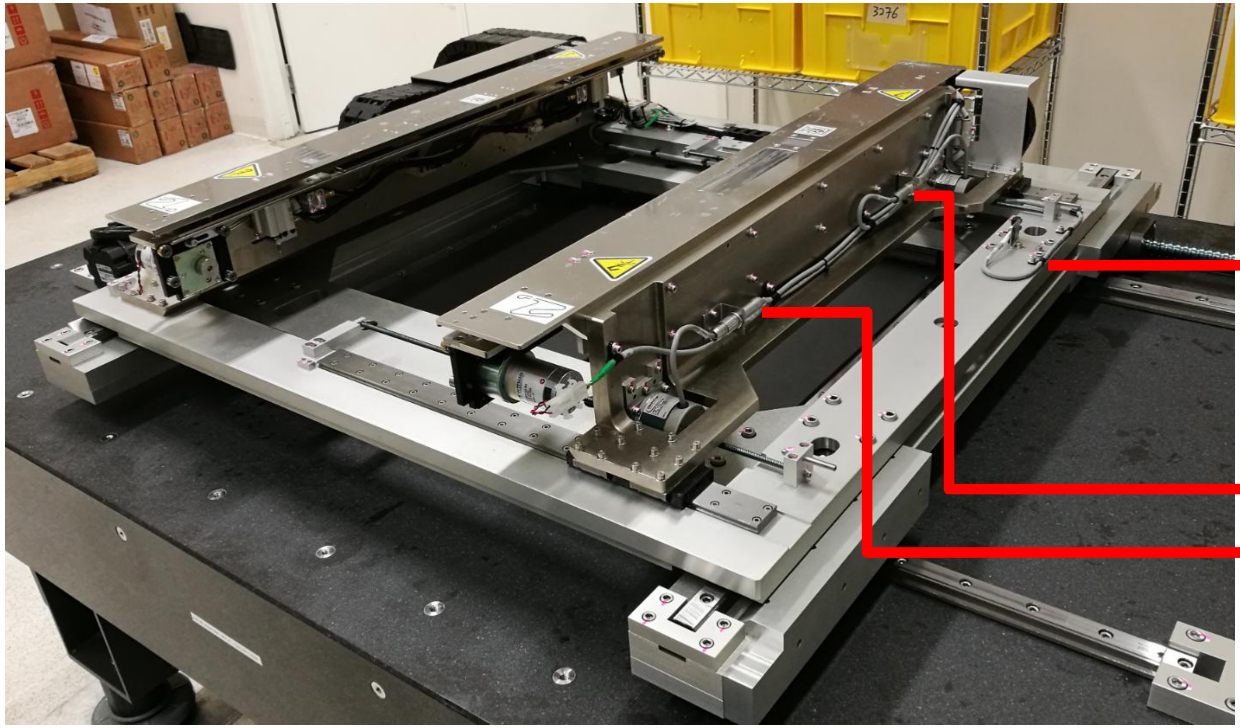

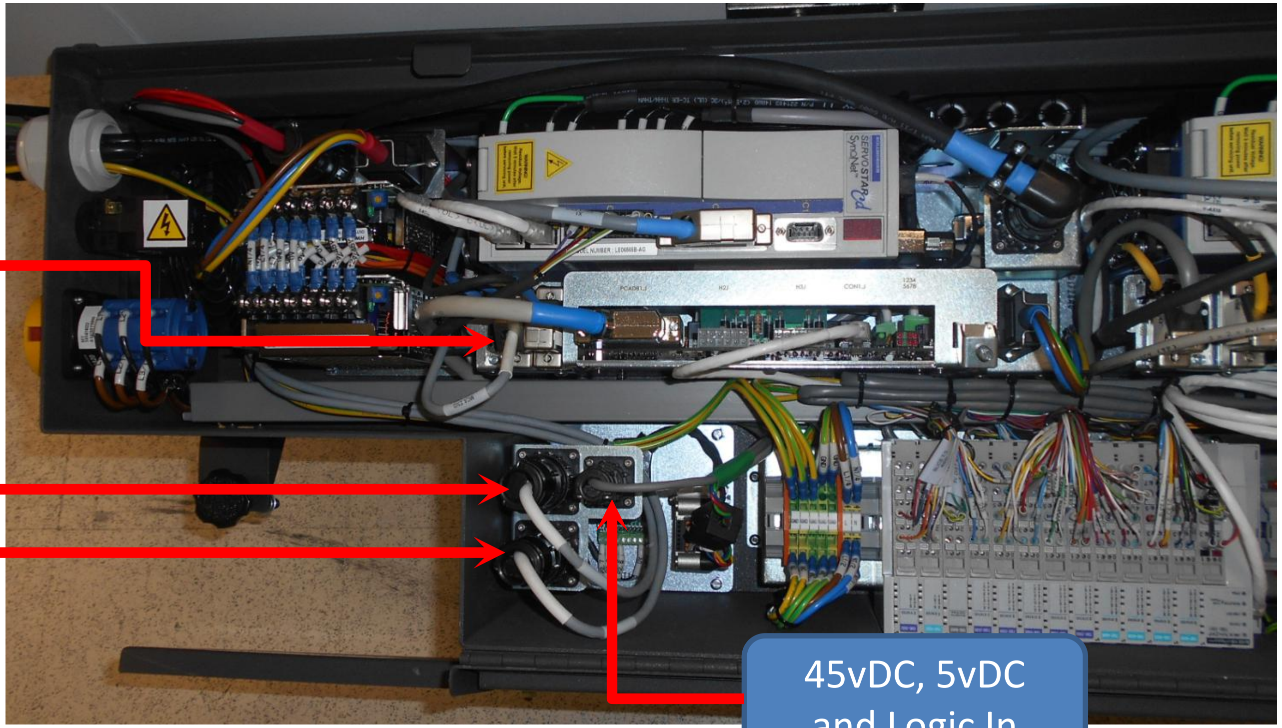

- Left and Right motor cable have no orientation as X-Koll assemble control rail width in parallel.
- 48vDC Lamba Power Supply (Configure as 45vDC) direct supply current to Stepper driver module
- 5vDC and Logic In supply from DIO Block

Hardware Overview - Auto Width Adjustable (AWA)

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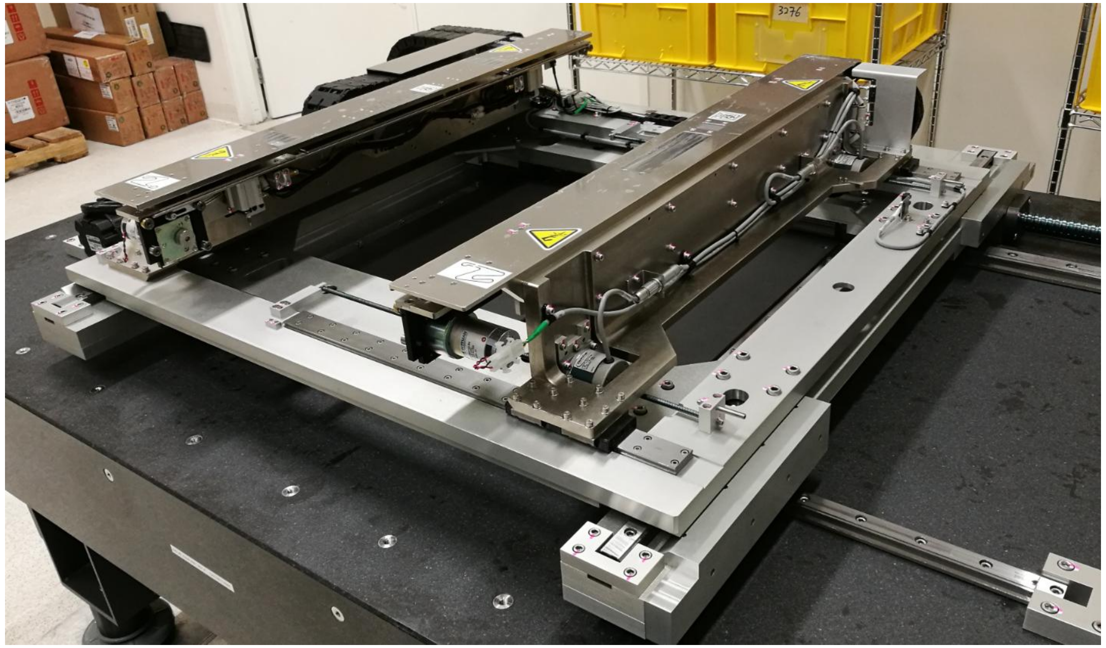

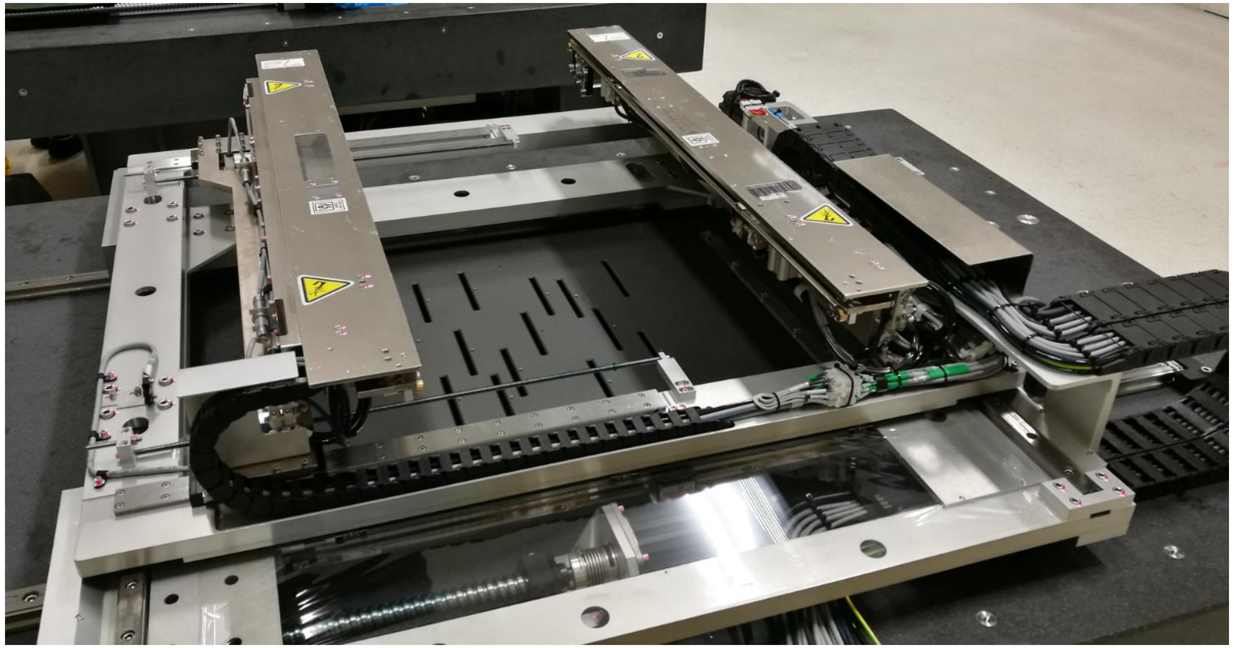

- Moves the Adjustable Rail to the correct distance from the Fixed Rail for the panel to be loaded into the system and inspected
- AWA have 2 stepper motor, each one at the Adjustable Rail end to drive the opening of the Adjustable Rail to correct width
- Have 2 lead screws
- 48 VDC supply is set to the both stepper motor

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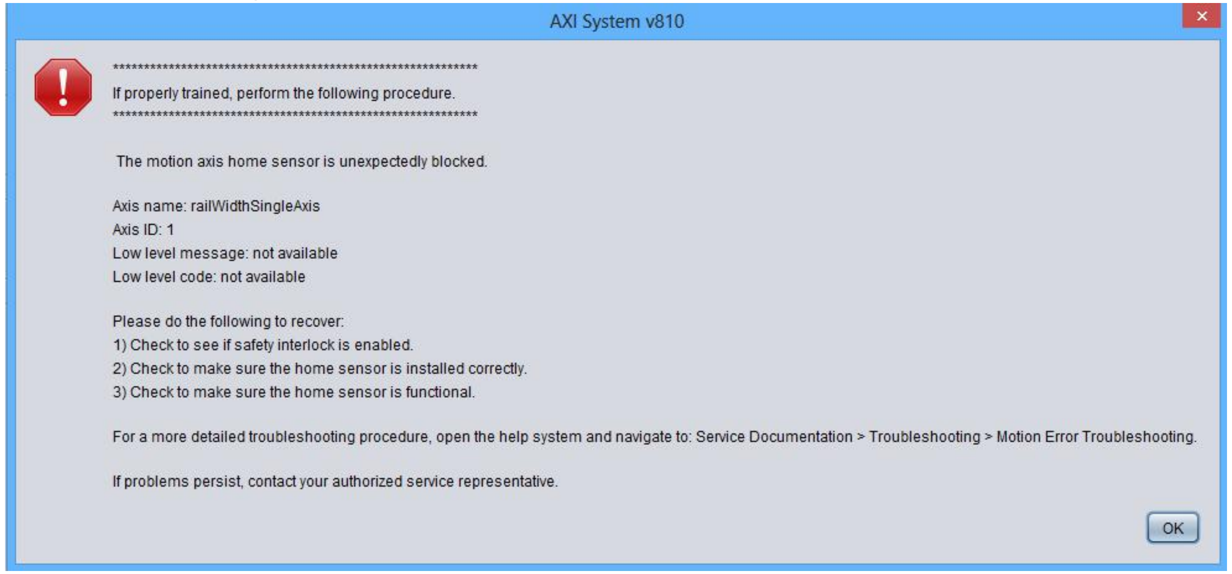
Troubleshooting - AWA Homing Issue



Potential Root Cause:

1. AWA home sensor misalignment/Mul-Function
2. AWA home sensor cable or Left or right AWA motor cable connectivity problem
3. X-Axis Kollmorgen Drive Kit, Servo interface card spoiled.

ABI Troubleshooting - AWA Homing Issue

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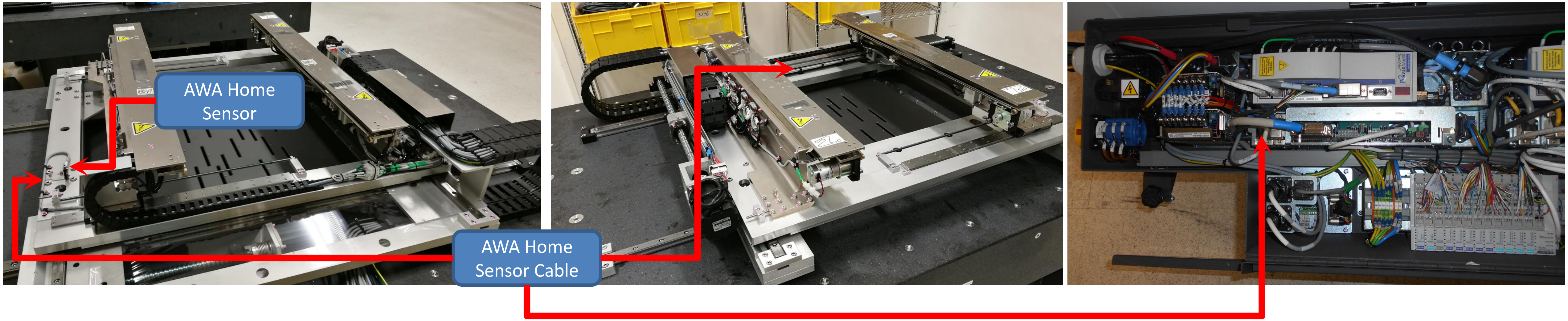
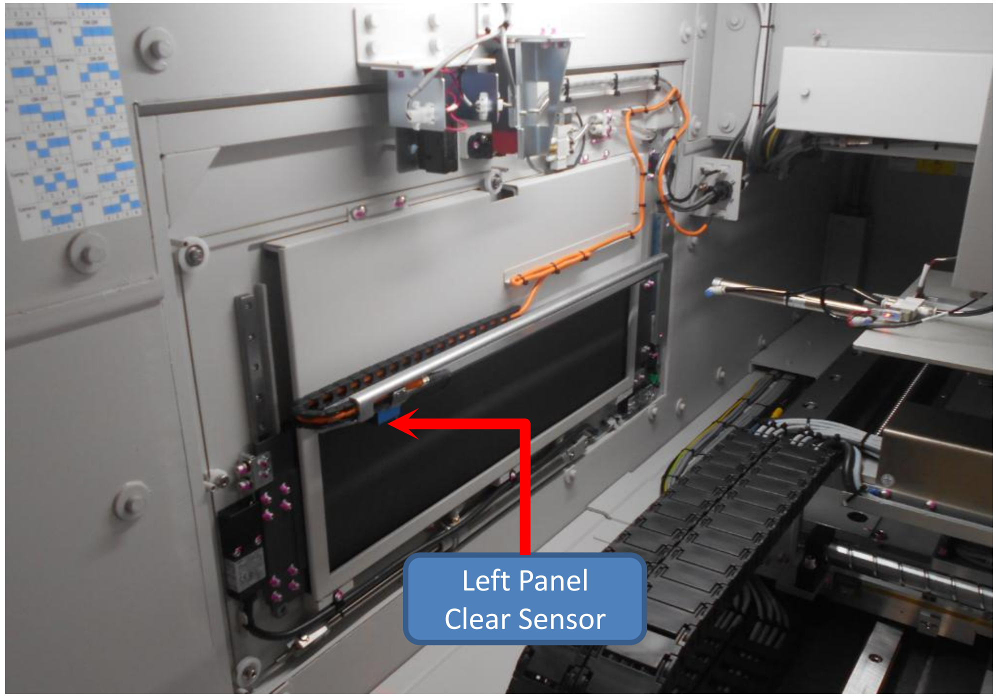
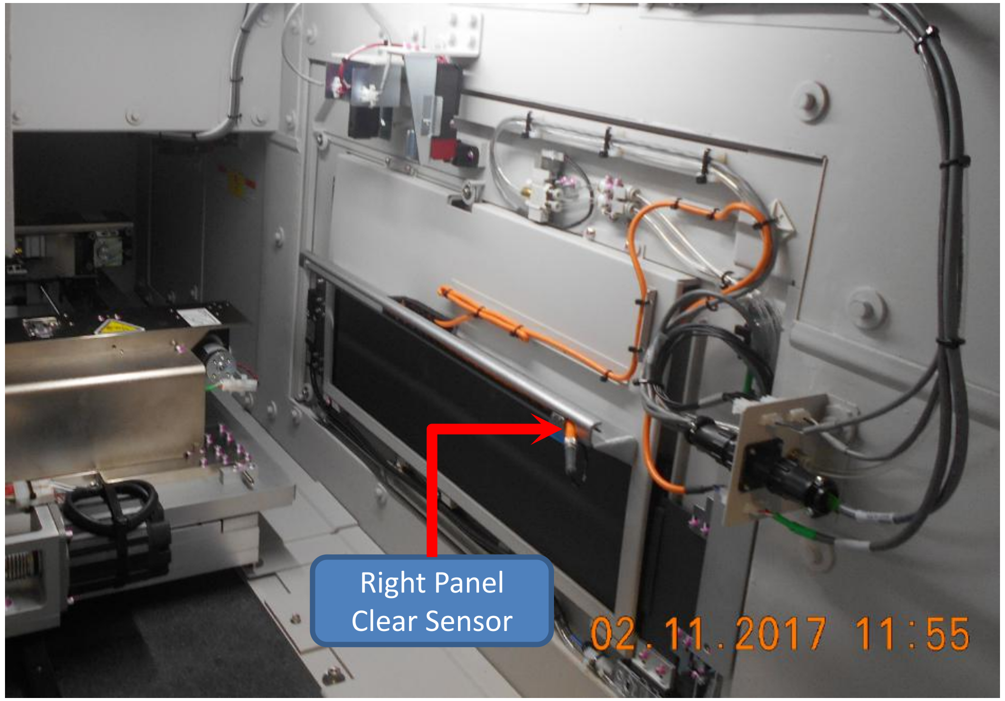


Troubleshooting Tips:

1. Remove and physical check on the home sensor and make sure no mirco crack on sensor or wrong P/N used.
2. Check the cables connectivity. May refer to remove and replace procedure for technical drawing
3. Visual check on cables to ensure no snapping issued
4. Reseat the connectivity for AWA home cable, both end.

Hardware Overview - Panel Clear Sensor





- To detect if any object present under sensor before XY table move
- To trigger conveyor belt on for board indexing
- S2ex machine type, Panel Clear Sensor did not have reflector
- Both left and right Panel Clear Sensor is mirror position
- The intensity/sensitivity of the Panel Clear Sensor is adjustable

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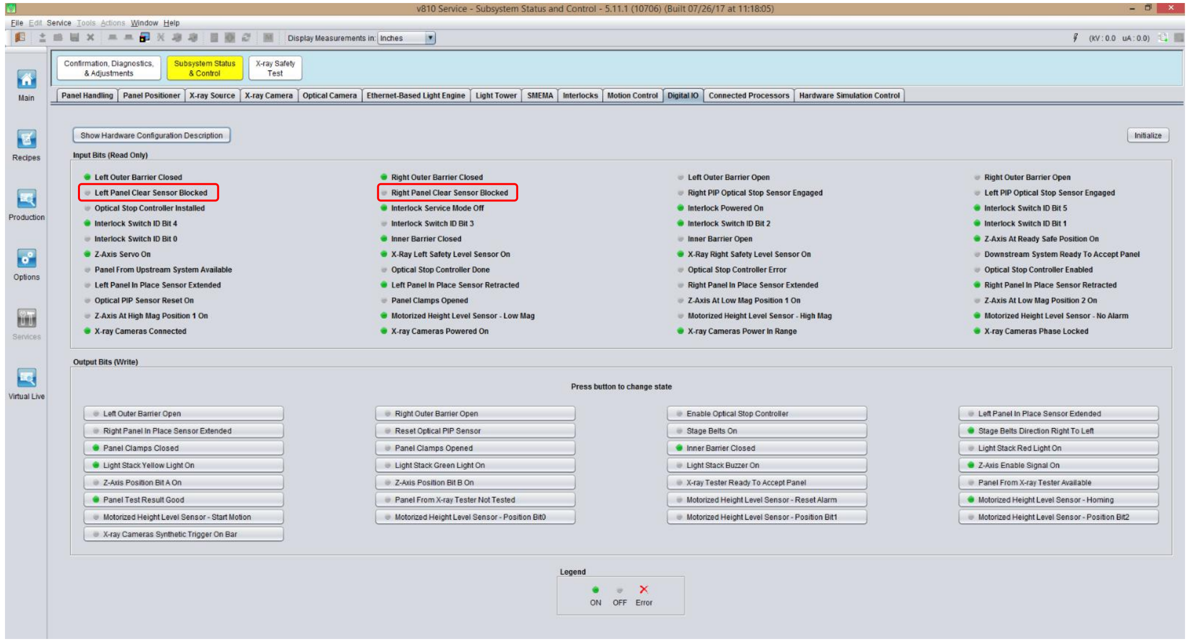
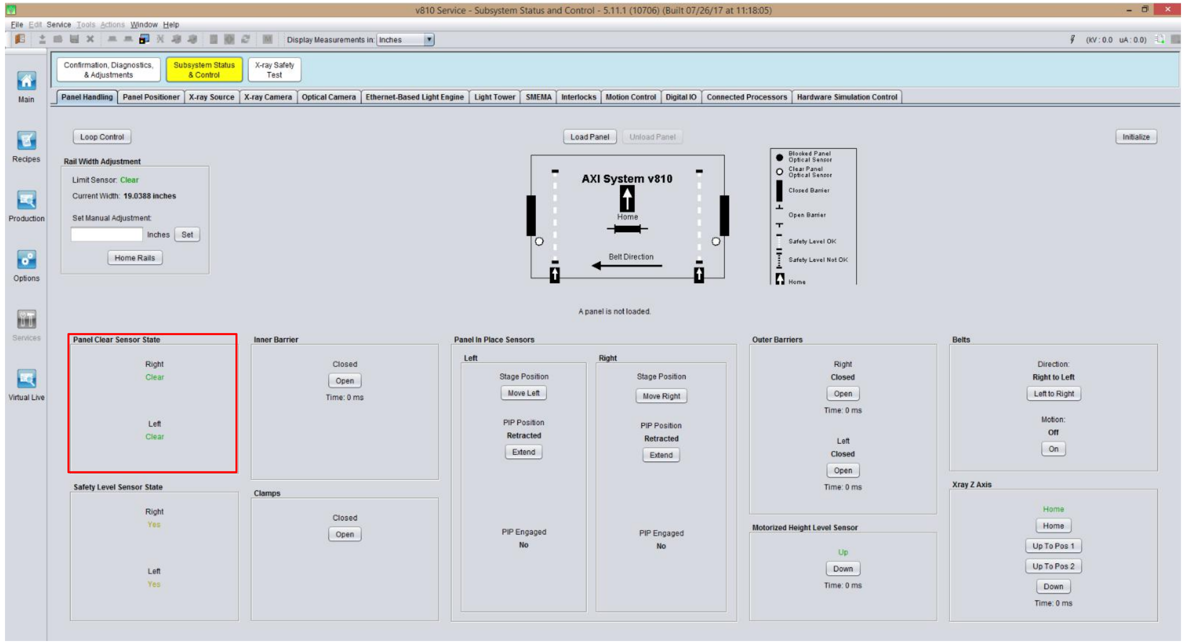
Hardware Overview - Panel Clear Sensor



DIO tab



Panel Handling tab

- Panel Clear Sensor should always in “Clear” status except board present on top conveyor belt surface (clamp or open).
- On Panel Handling tab, if Panel Clear sensor in “Blocked status” (Green Light), XY table unable to Home Stage.

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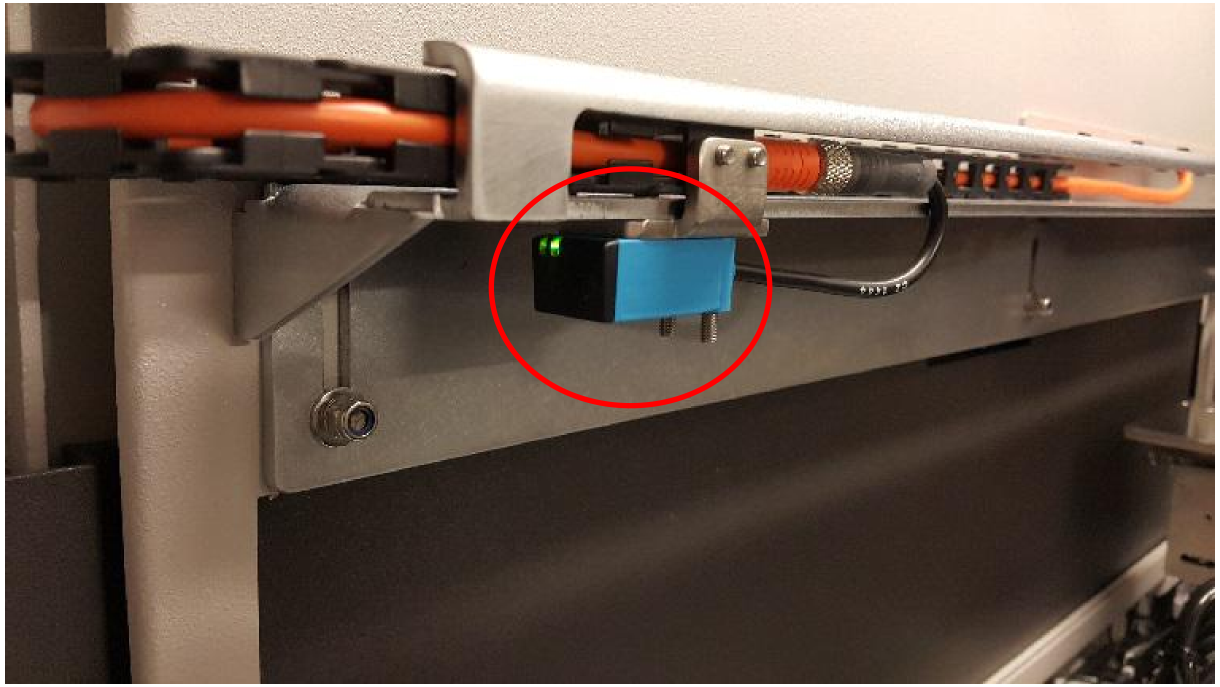
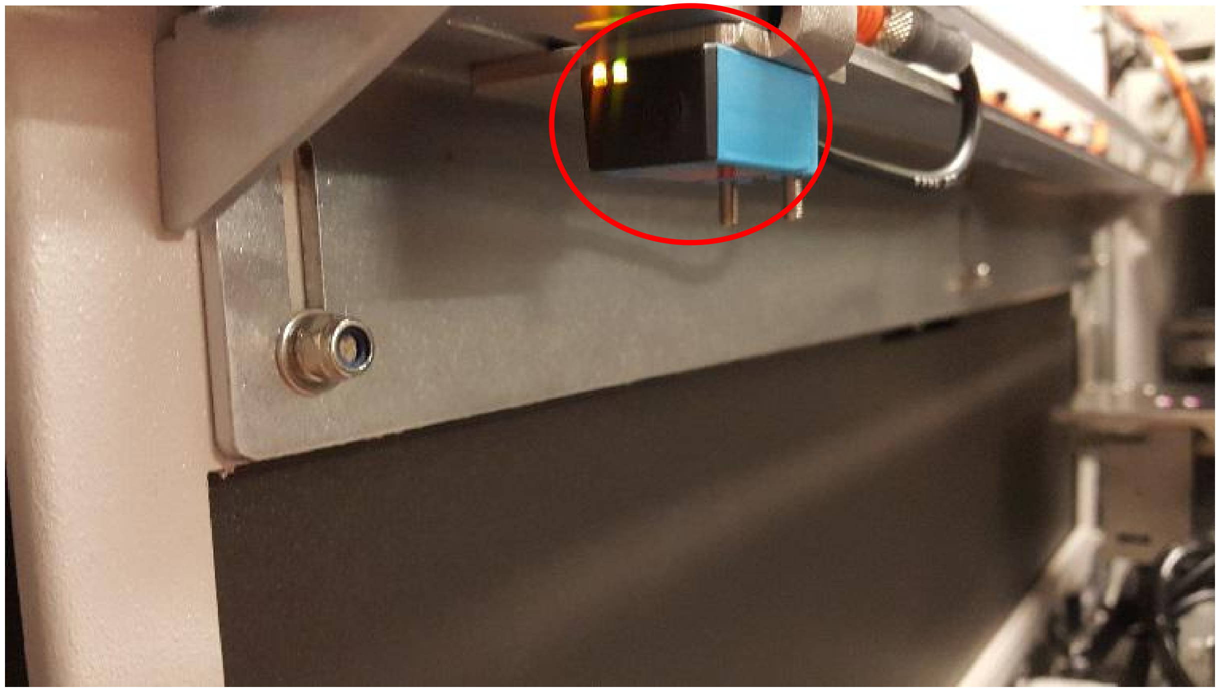
Hardware Overview - Panel Clear Sensor



Panel Clear Sensor Clear Status



Panel Clear Sensor Blocked Status

Hardware Overview - Hybrid Panel In Place Sensor

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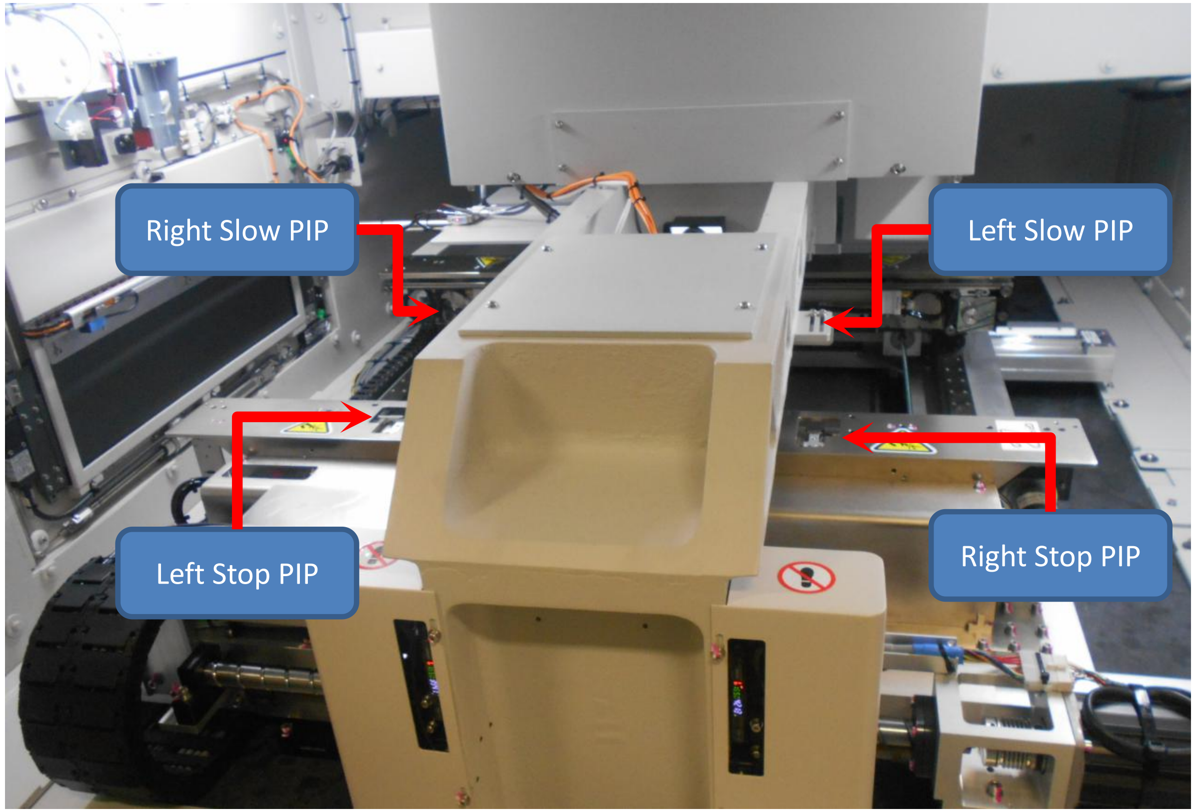

What is Panel In Place (PIP) Module?

- PIP serve as the module to stop the board at the accurate stopping position while loading the board
- Hybrid PIP (Optical Slow + Mechanical Stop) is used to replace the Rail Optical PIP to serve the same purpose.
- To trigger board stop after board indexing complete in position

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Hardware Overview - XY Stage - Panel In Place Sensor

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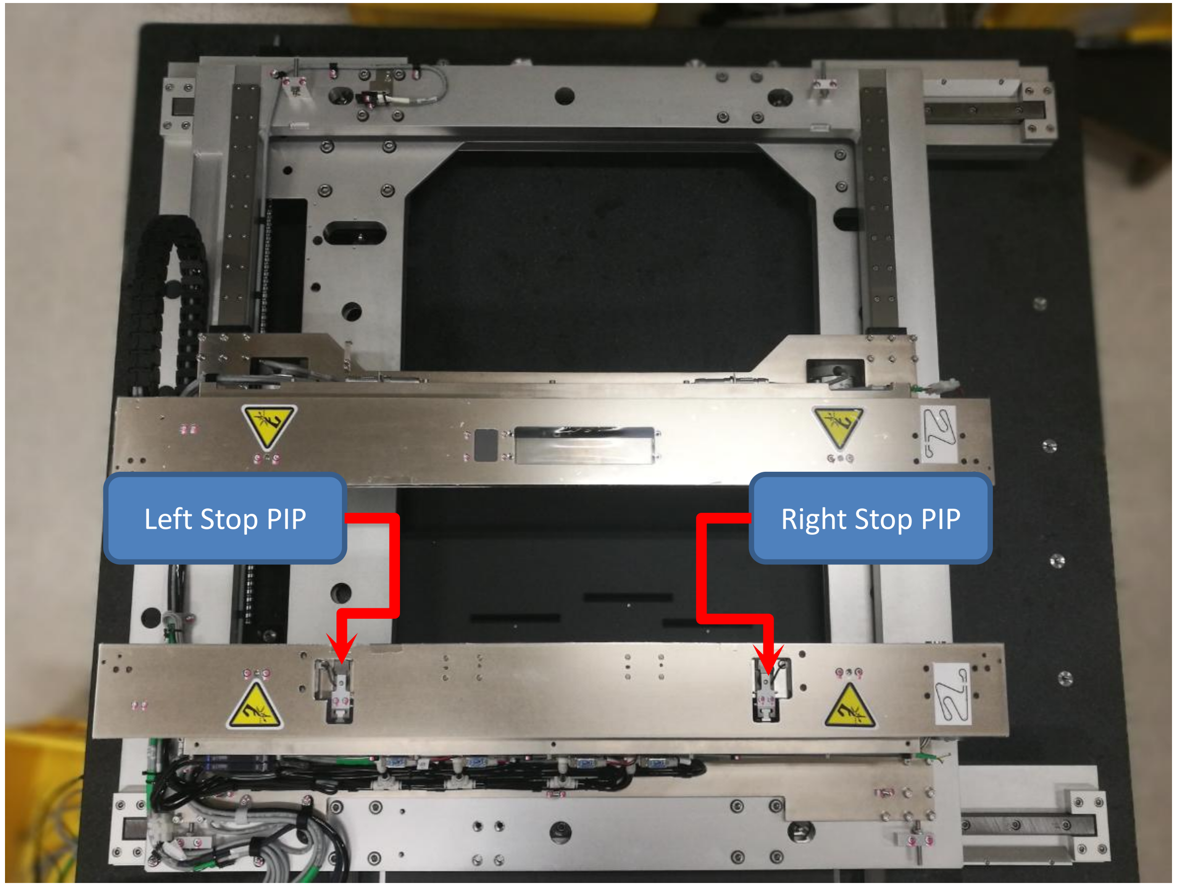

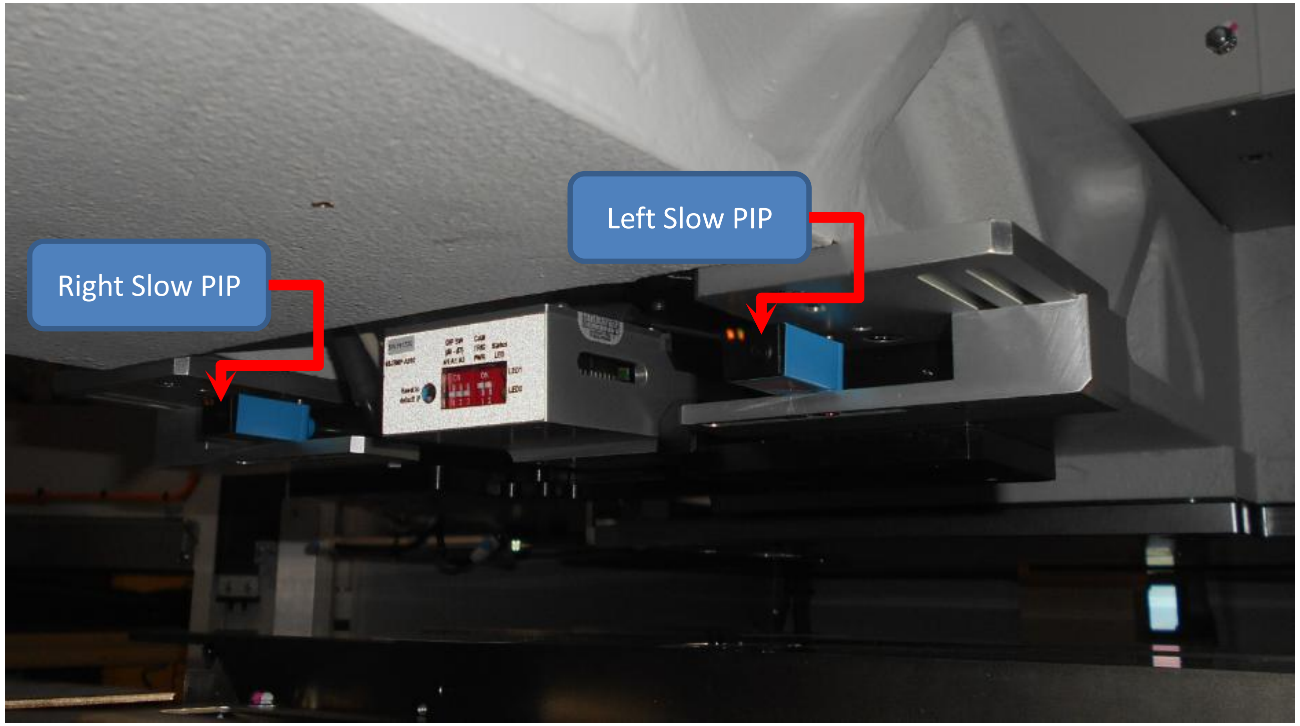

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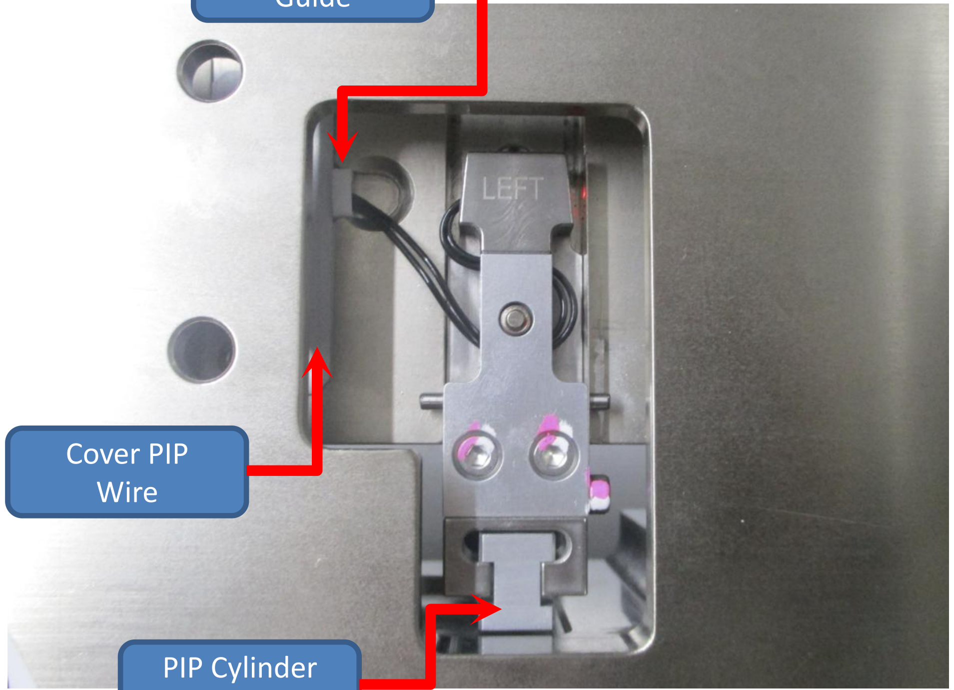
Hardware Overview - Hybrid Panel In Place Sensor

Teflon PIP Guide

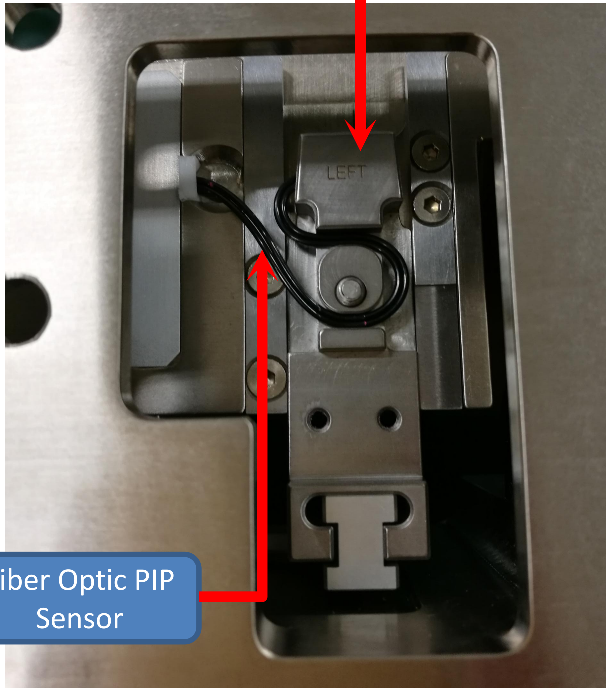
Cover PIP Wire

PIP Cylinder Nose



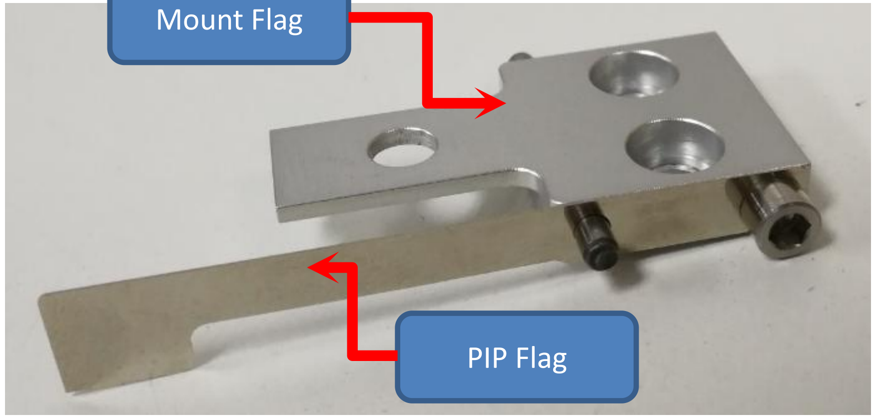
Board Stop Mount

Fiber Optic PIP Sensor



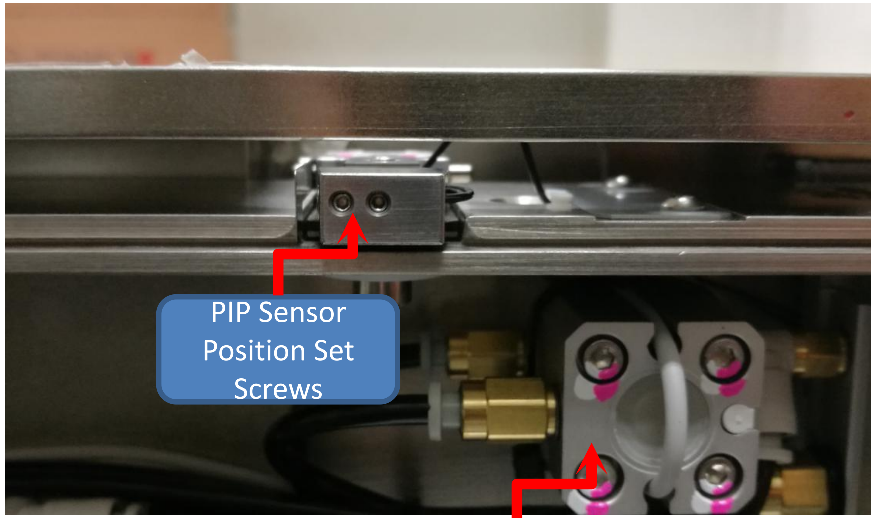
Mount Flag

PIP Flag

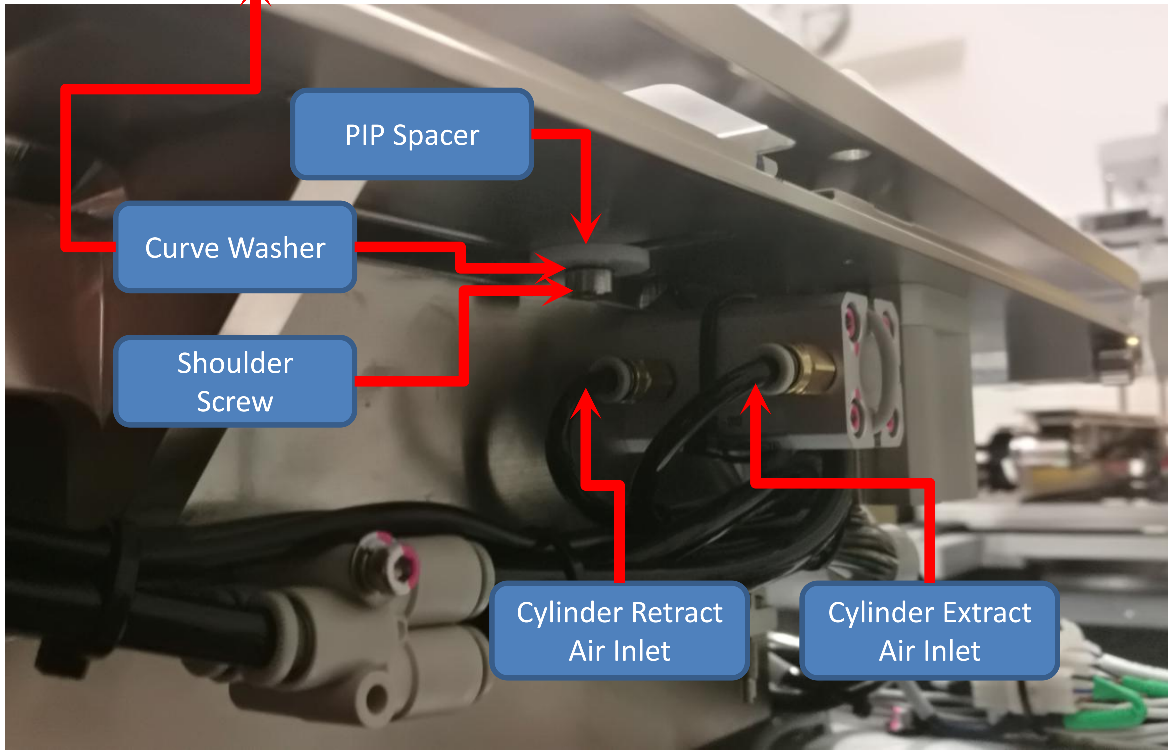


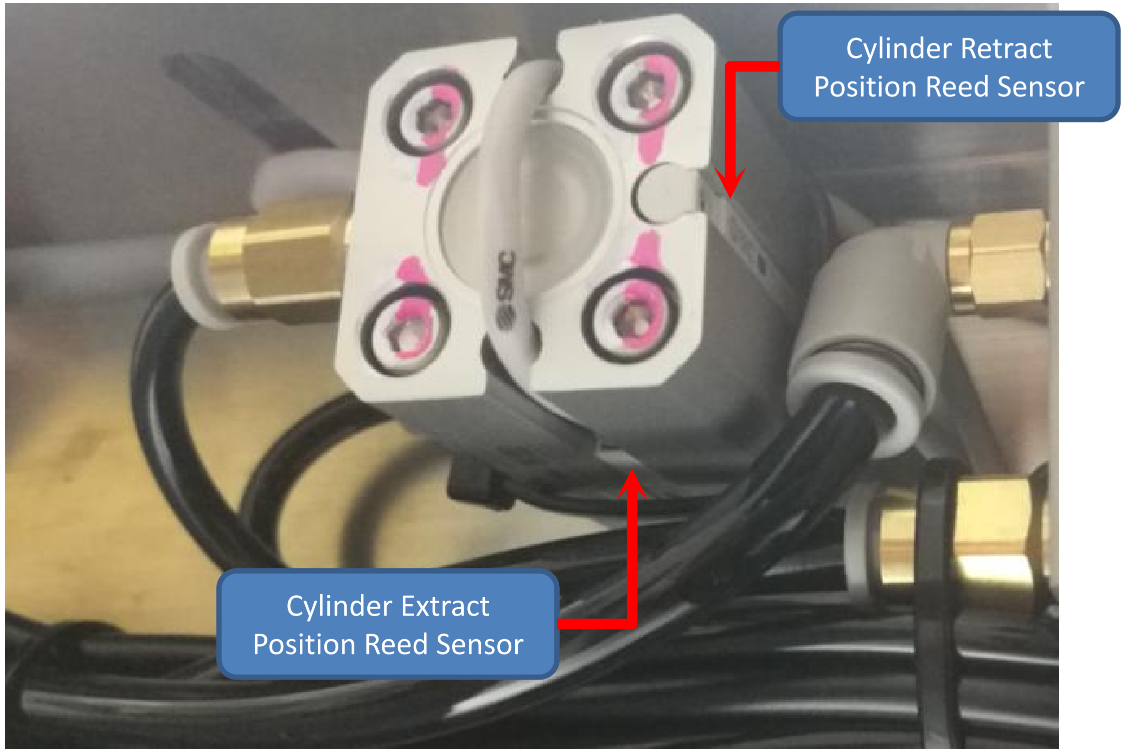
PIP Sensor Position Set Screws

PIP Cylinder



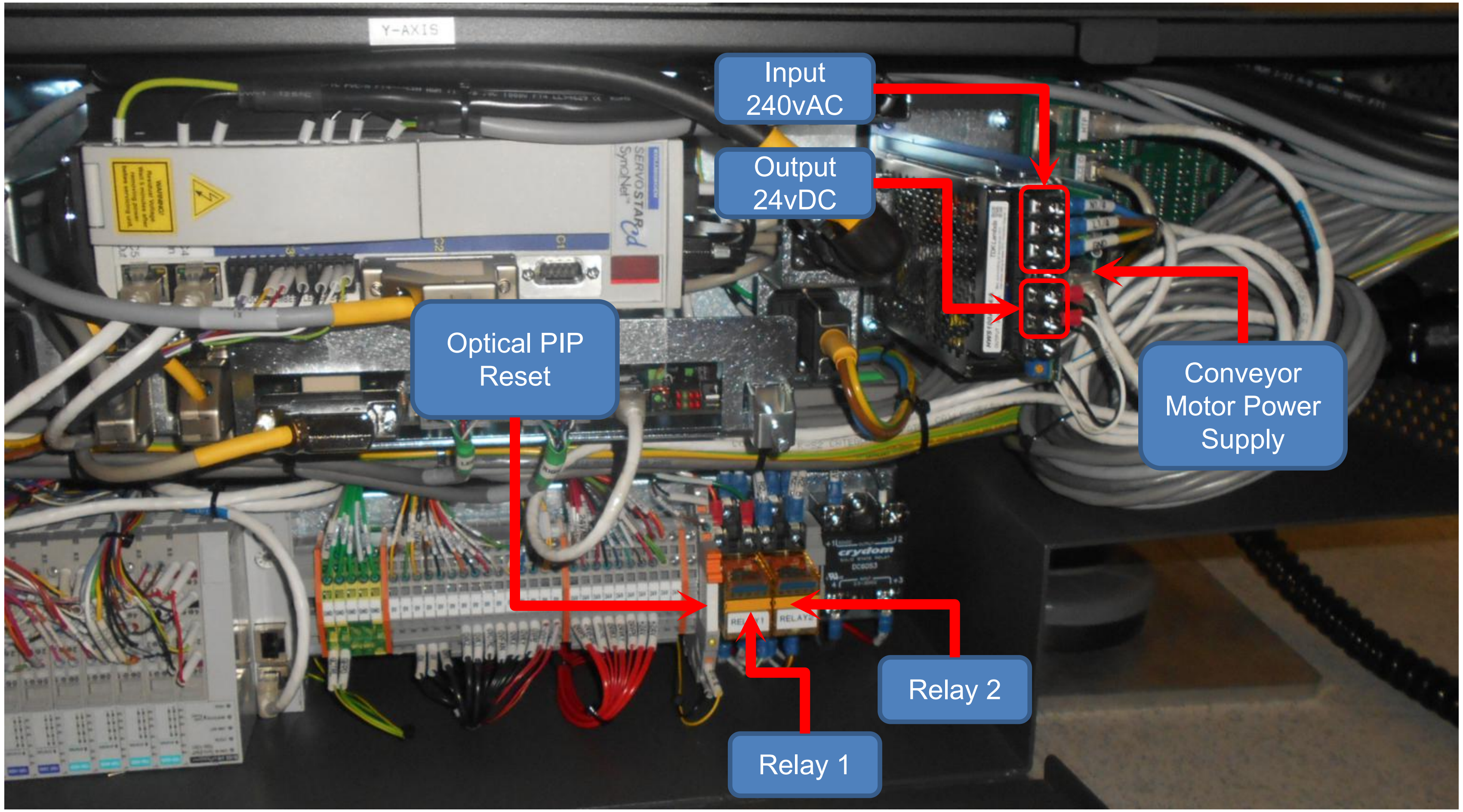
Hardware Overview - Hybrid Panel In Place Sensor





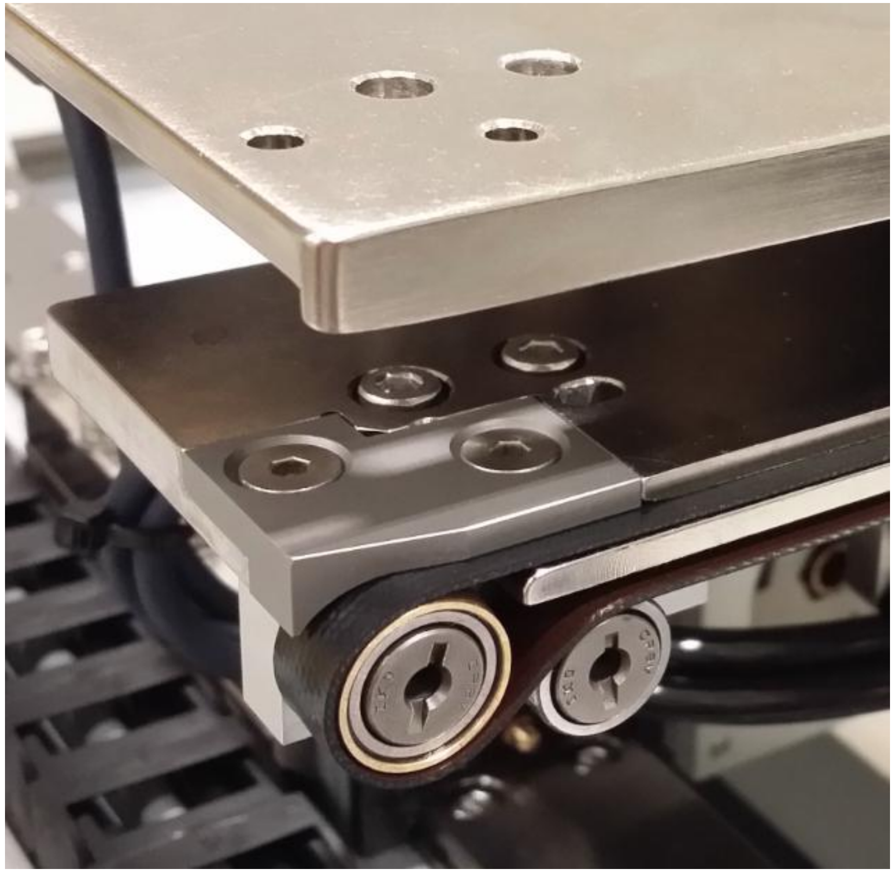
ABI Hardware Overview - Belts, Motors and Relays

Automated Board Inspection

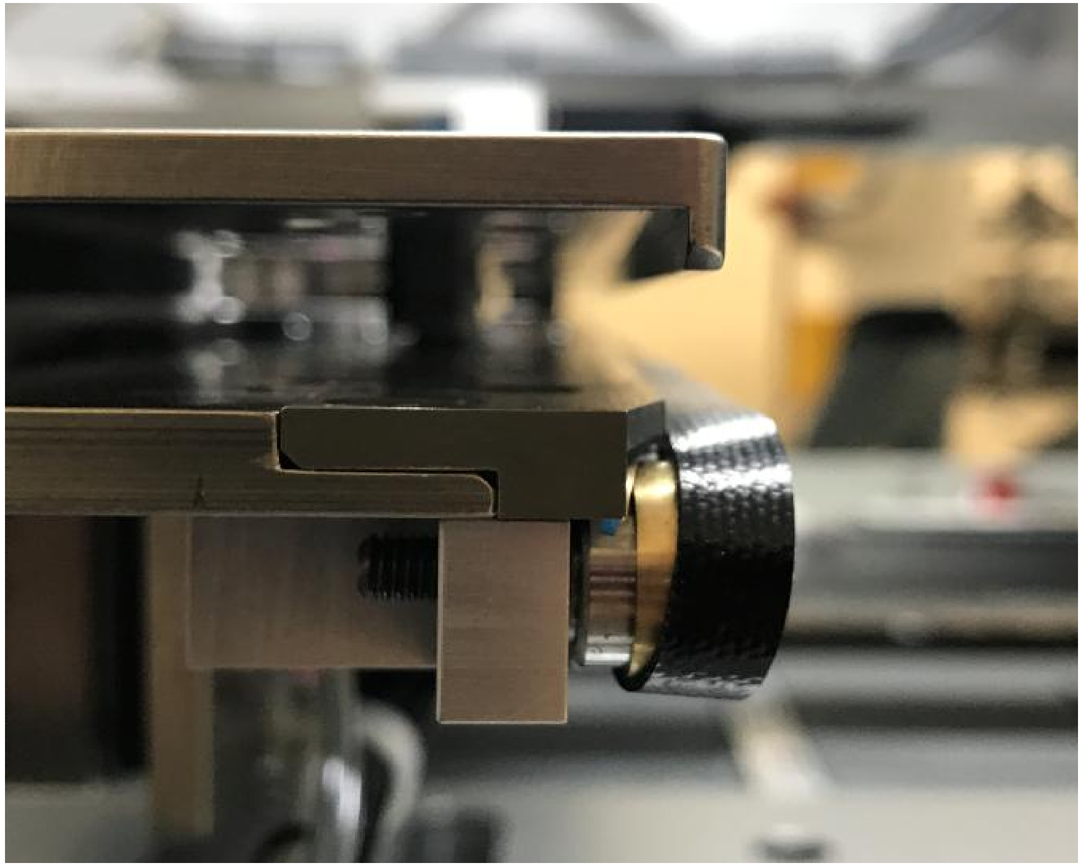


- Relay 1 = On/Off Conveyor Motor
- Relay 2 = Conveyor belt direction change (forward/revert)
- Optical PIP Reset Relay = Reset PIP Amplifier (Only applicable to Rail Optical PIP)
- Conveyor Motor Power Supply (24vDC) = supply current to both conveyor motor

Hardware Overview - Conveyor Belt Alignment



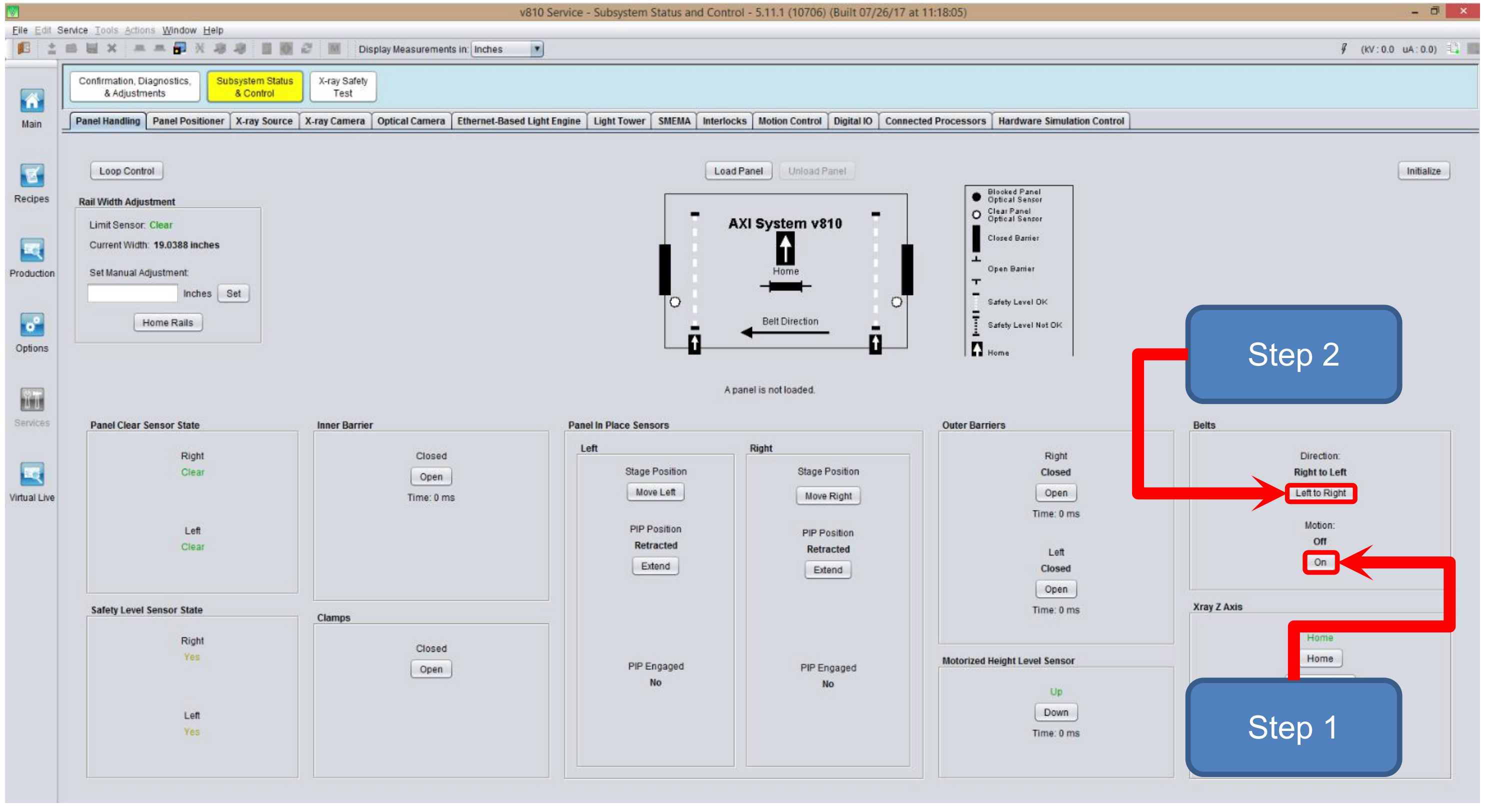
Belt in alignment

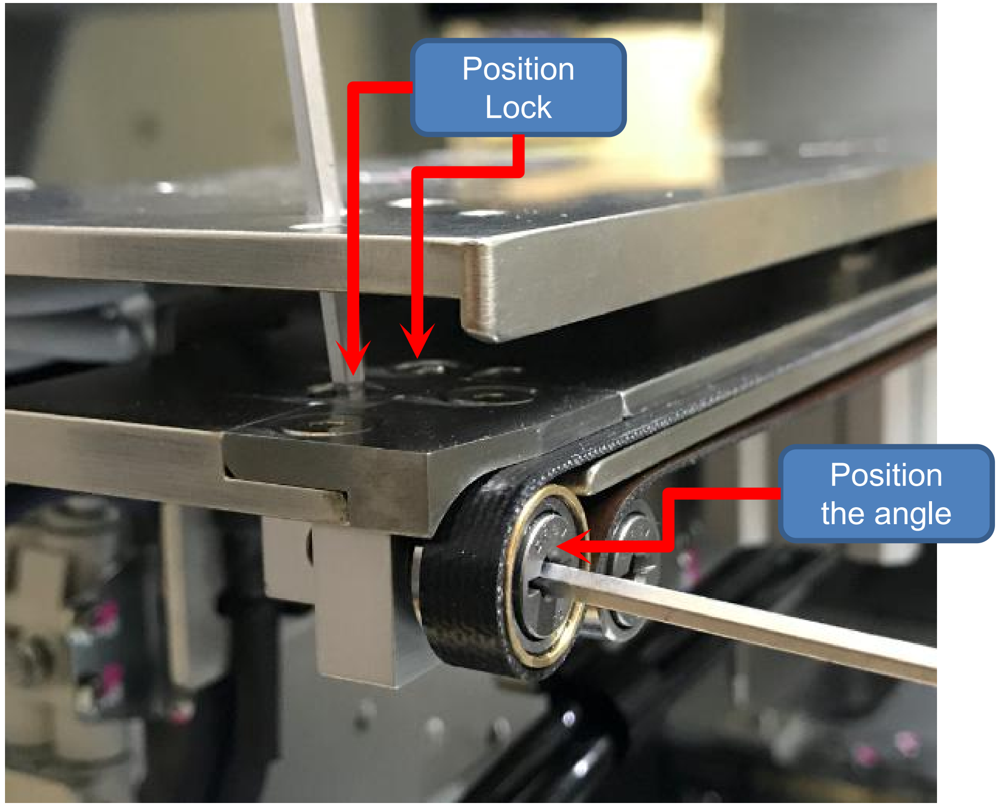


Belt not in alignment

Hardware Overview - Conveyor Belt Alignment

ABI
Automated Board Inspection





1. Turn the conveyor belt on
2. Reposition the End Bearing Block until the belt is spinning middle of the pulley kit and tighten the screws for position lock.
3. Run belt in forward and revert direction and validate the belt alignment

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